The emergence of grassroots activism through microblogging: Tweeting the oil spill

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Abstract
This paper examines the emergence of grassroots activism through microblogging, i.e. through the writing and broadcasting of very short messages online via services such as Twitter. Grassroots activism corresponds to symbolic action dedicated to changing the status quo of a situation that is organized and coordinated from the bottom up. We start by defining key concepts associated with the emergence of grassroots activism based on new social movement theory, on a multilevel theorization of emergence, and on the features and affordances of microblogging. We then examine a large and longitudinal set of microblog posts dealing with the Gulf of Mexico oil spill in order to develop a grounded theory of emergence of grassroots activism. Our findings reveal the role of microblogging in four key dimensions of this process: information sharing, framing, community building, and mobilizing. We then present a grounded theorization of grassroots activism emergence that shows how these dimensions progressed and were related to each other and to features and affordances of microblogging. We draw implications of this research regarding the significance of grassroots activism through microblogging and the fluidity or persistence of such phenomena. We conclude by summarizing our contributions to IS research, acknowledging the limits of our work, and suggesting avenues for future research.

Key words: Symbolic action, affordances, social media, environmentalism, multilevel theory, framing, new social movement.

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"Sure, hashtags come and go, and the so-called weak ties of digital movements are no match for real world engagement. But they are not only better than nothing, they probably make the world, the one beyond the keyboard, a better place." David Carr, 2012, *New York Times*, March 25.

INTRODUCTION

Wattal et al. (2010 p. 680) recently stated that "*the field of Information Systems can move beyond the traditional focus on business and the organization and tackle larger societal issues*." Such an assertion has taken on special significance for Information Systems (IS) scholars given that new web-based applications of the "web 2.0" generation have disrupted established practices in business and organizations *and* have contributed to intriguing societal transformations. For instance, the ways in which people share and comment upon news, ranging from technology to sports, fashion, celebrity or political events, have been altered by new tools such as social networking, blogging, or microblogging (Davidson & Vaast, 2009; Dhar & Chang, 2009; Wattal et al., 2010). Microblogging, an increasingly popular type of social media that involves the broadcasting of very short messages through the Internet, has in particular recently been associated with new forms of social engagement and with social uprisings (Andrew, 2009; Fleishman, 2009; Shirky, 2011; Solow-Niederman, 2010). People's growing reliance upon microblogging not only to talk about but also to act upon specific situations and events has been documented in contexts such as the protests following the 2009 Iran elections (Gaffney, 2010; Morozov, 2009), uprisings of the "Arab Spring" (Lotan et al., 2011; Maghrabi & Salam, 2011), or of the organization of environmental awareness events (Cheong & Lee, 2010).

What is striking about these examples is not only that microblogging has fast become what seems to be an "inevitable" tool for activists but also, and more theoretically puzzling, that microblogging seems to enable the emergence of grassroots activism, i.e. of collective activism that is not organized or spearheaded by a formal organization or clear leader but, rather, that

comes from the bottom up (Franklin, 2007; Fuchs, 2006; Neumayer & Raffl, 2008). In this research, we thus examine the following research question: *How does grassroots activism emerge through microblogging?*

Such question is topically relevant, as illustrated by the aforementioned examples; it is also theoretically important. This question deals with and forces scholars to revisit complex issues regarding the relationships between the individual and the collective and the production of symbols and of action through media (Adorno, 1991; Austin, 1962; Clark, 1996; Habermas, 1984). Habermas (2006) provided in this regard a seminal outlook of the consequences of the Internet and social media on the public sphere in democratic societies:

> In the context of liberal regimes, the rise of millions of fragmented chat rooms across the world tends instead to lead to the fragmentation of large but politically focused mass audiences into a huge number of isolated publics. (Habermas, 2006: 423).

This nuanced assessment goes to the core of our research question because it questions whether and how "*isolated publics*" might coalesce in a bottom-up fashion towards a collective. It also implicitly inquires whether "*fragmented chat rooms*" – and, one would add, microblogging communities – might turn from interested publics into grassroots activists. Social and political scientists have debated whether microblogging actually contributes to effective activism (Breindl & Francq, 2008; Gladwell, 2010; Shirky, 2011; Van Laer & Van Aelst, 2010), but they have neither theorized the roles that microblogging actually plays nor the process of emergence itself.

We believe that the IS discipline has a lot to say about *and* to gain by tackling this issue. Wattal et al. (2010) who propounded that IS scholars "*tackle larger societal issues*" (p. 680) went on to say that the IS field is especially useful for its "*two unique perspectives*" of being "*process-centric*" and of "*comparison of technologies and matching "features" to behaviors and*

outcomes" (p. 680). IS theory and research is therefore promising to understand better how symbolic action enabled by social media such as microblogging develops and transforms. This paper examines the emergence of one such form of symbolic action, grassroots activism.

The IS field also has much to benefit from examining such issues. Even if these issues have not yet become part of the core of our discipline, they reflect ongoing and current crucial changes in organizing, coordinating, innovating, value creating, etc. that are enabled and constrained by web 2.0 applications (Hansen et al., 2009; Kane & Fichman, 2009; Ransbotham & Kane, 2011). The IS discipline has started to develop conceptual and methodological resources to help make sense of these fundamental new changes in our social, business, and political lives (Faraj et al., 2011; Franklin, 2007; Shirky, 2009; Wattal et al., 2010). Managers, decision makers, citizens have become aware of these ongoing transformations but much is still unknown regarding their meaning and implications (Fuchs, 2006). IS research should tackle such issues to remain relevant in these changing technological and organizational times as well as to maintain and enhance its academic "*legitimacy*" by adjusting its methods and studying such important new issues to produce new theories (Lyytinen & King, 2006).

This research constitutes a step in this direction and makes four contributions to IS research. First, it builds a grounded theorization of grassroots activism through microblogging that responds to Wattal et al. (2010) and Lyytinen and King's (2006) call for IS research to enhance its standing by examining critically key trends and transformations in organizational, economical and political contexts. Second, it builds a model that is multilevel and that encompasses emergence on a very large scale, which is suited to the theorization of dynamics associated with the use of social media. Third, it develops a process theory that takes in consideration how features and affordances of social media combine in time to shape grassroots

activism. Finally, methodologically, this paper represents one of the first attempts to tap into the huge wealth of social media data to generate theory rather than to describe phenomena or to test existing theories.

In the remainder of this paper we first present the theoretical foundations of our work to define conceptually the main concepts that we examined. We then turn to our empirical investigations of a large and longitudinal set of microblog posts related to the Gulf of Mexico oil spill. We present how we triangulated data and methods in order to develop a grounded theory of emergence of grassroots activism through microblogging. We then present implications of our work regarding the significance of microblogging-enabled activism and the persistence over time of such phenomena. As we conclude, we acknowledge the limitations of our work and provide avenues for further investigations.

THEORETICAL FOUNDATIONS

This section defines key concepts that informed our research based on new social movement theory, on multilevel theory, and on a perspective of the features and affordances of microblogging.

Activism

New Social movement (NSM) theory has conceptualized social movements and activism by considering not only their political dimensions but also their social and cultural aspects (Buechler, 1995; Morris & McClurg Mueller, 1992; Pichardo, 1997). According to NSM theory, activism aims at changing power relations and at improving upon an existing situation to correct injustice, wrongdoing, or disasters. Activism "*involves attempts to change the status quo, including targets such as social norms, embedded practices, policies, or the dominance of certain social groups*" (Zoller, 2005, p. 344).

Activism supposes symbolic action with its intricate relationships between action and discourse through the production of symbols. Activists create symbolic resources and engage in symbolic action to set right things they perceive as wrong (Burke, 1978). In this paper we rely upon Burke's (1966) definition of symbolic action: "*Language is a species of action, symbolic action--and its nature is such that it can be used as a tool*." (p. 15). Common activist tactics illustrate the importance of symbolic action and of the role of language used as a tool to change the status quo. Such tactics include the organization of and participation in protests, public awareness and advocacy campaign as well as legal and negotiation strategies (Ford & Yep, 2003; Snow et al., 2004).

Activism can develop at the individual or collective level (Zoller, 2005). Individual activism is carried out by individuals or isolated entities (e.g. a local organization) and does not require social organization and coordination (Zoller, 2005). By contrast, collective activism is a form of social movement that relies upon entities coming together to pursue a common goal (Della Porta & Diani, 1999; Zoller, 2005). Collective activism involves formal and informal networks, shared beliefs, solidarity, and collective action (Snow et al., 2004). Collective activism differs from isolated protest events or entities and involves a shared vision, identity, interrelationships, and a sense of ongoing joint action (Friedman & McAdam, 1992).

Because of its close links to collective action (Olson, 1971), collective activism has traditionally relied upon the involvement of a very active minority recognized for its role in the advancement of the cause (Morris & McClurg Mueller, 1992; Moscovici, 1979). Examples of such active minorities include non-profit organizations, political and rebel groups. This traditionally dominant form of collective activism can be qualified as "top-down:" a recognized, more or less formal, entity incurs most of the cost of engaging in the collective action on behalf

of the collective, with the expectation that other members of the collective will follow in time to achieve the activist mission (Zoller, 2005).

Grassroots activism, which can be defined as *"a form of collective advocacy in behalf of a shared cause or direct action in the service of achieving a collective goal"* (Wittig, 1996, p. 4) constitutes a second, alternative, model of collective activism. Grassroots activism has been of special interest to NSM theory as it relies on a bottom up process of organizing where members "*are not appointed, elected or recruited except by themselves*" (Zander, 1990, p. 22). It implies a process with lesser emphasis on the role of formal entities, where new knowledge emerges from intergroup, informal relationships. Table 1 contrasts individual, top-down, and grassroots activism.

-- Insert Table 1 around here --

Our interest in this paper is on examining the process of emergence from individual to grassroots activism. Building upon seminal research on social movements and IS and social media (Carroll, 2006; Diani, 2000; Kahn & Kellner, 2004), we believe that social media such as microblogging can enable and shape the emergence of grassroots activism. We thus now turn to a definition of emergence and discuss key features and affordances of microblogging.

A multilevel perspective on emergence

In this paper we view the emergence of grassroots activism as involving multiple, initially isolated and fragmented entities gradually coming together around a shared cause and creating an activist collective. Multilevel research provides us with a theoretical perspective to conceptualize such emergence.

Multilevel research straddles the boundaries between individuals, groups, and organizations *"to bridge the micro-macro divide"* (Klein et al., 1999, p. 243) and to allow for a

"*more integrated understanding of phenomena that unfold across levels in organizations*" (Klein & Kozlowski, 2000, p. 7). Seminal IS research has also advocated for and applied multilevel analyses and theorizing (Burton Jones & Gallivan, 2007; Lapointe & Rivard, 2005). Multilevel perspectives provide a theoretical framework and methodological toolbox that help organization and IS researchers theorize and operationalize constructs at various levels as well as the relationships among these levels (Klein et al., 1999). Multilevel perspectives are thus essential to understand, in dynamic, how higher-level phenomena *emerge* from lower-level constituents, which conceptually echoes the progression from individual to grassroots activism.

The emergence process, which by definition is bottom-up, is shaped by properties inherent in the lower-level components as well as by the interactions that exist between these constituent elements. Morgeson and Hofmann (1999) conceptualize emergence as coming from interactions among multiple entities and resulting in a "*structure of collective action* (…) *that transcends the individuals who constitute the collective*" (p. 252-253). (Kozlowski & Klein, 2000) (p. 55) similarly consider that:

> A phenomenon is emergent when it originates in the cognition, affect, behaviors, or other characteristics of individuals, is amplified by their interactions, and manifests as a higher-level collective phenomenon.

Further untangling emergence, (Kozlowski & Klein, 2000) conceptualize collective phenomena as enacted from *composition* or *compilation* of lower level constituents onto the collective. Compilation assumes heterogeneity in lower-level characteristics and involvement, and involves non-linear combination of these diverse lower-level elements to see the emergence of distinct patterned subsets of higher-level collective phenomenon. Composition, on the other hand, assumes homogeneity in lower-level characteristics that converge, in an almost linear fashion, to yield a higher-level phenomenon or behavior that is essentially the same as its

constituent elements. Through emergence, as entities come together from the individual to the collective and become closer to one another, there might be a transition from compilation to composition (Kozlowski & Klein, 2000, p. 60 et f.).

Multilevel theorization also emphasizes that the emergence of collective phenomena from lower-level constituents manifests itself over time (Kozlowski & Klein, 2000). Examining emergence therefore involves adopting a temporal perspective and considering how lower-level constituents coalesce (or not) into a collective phenomenon over time.

Adopting such theoretical and methodological multilevel lens in the context of this study illuminates the process going from individual to grassroots activism. As extant literature on multilevel research has shown, individuals aiming to pursue a particular course of action often congregate in collectives in order to achieve critical mass and be able to influence and shape outcomes that are more in-line with their own views and priorities (Lapointe & Rivard, 2005). Consistent with insights from NSM theory, the concept of *emergence* of grassroots activism embodies this very idea of individual entities getting together in a non-centrally coordinated – but cohesive – way in order to change the status quo and achieve a desired goal.

Microblogging features and affordances

Given the objective of our study, considering the features and affordances of technological innovations such as microblogging is of paramount importance. Building on DeSanctis and Poole's (1994) concept of *structural features*, Markus and Silver (2008) conceptualize features as *technical objects*. As such, they denote *"IT artifacts and their component parts"* (pg. 620) as representing material things that exist independently of users' perceptions of them. The intrinsic properties that are embodied in these artifacts are the necessary and sufficient conditions that determine how they will be used and the impacts that they will have.

Gibson (1977) defined affordances as all *"action possibilities"* latent in the environment, independent of individuals' abilities to recognize them, while at the same time dependent on the actors' capabilities. The concept has been introduced into the organization and IS fields and has shown that *"at its core, an affordance perspective recognizes that a technological object has some recognized functionality but needs to be recognized as a social object"* (Zammuto et al., 2007, p. 753). As such, functional affordances are viewed as *relations* between the technical object and its user (Markus & Silver, 2008), rather than as an intrinsic property of the technical object. Because of this sociomaterial component, the way these objects are appropriated and their impacts are not deterministic in nature but are rather shaped through human agency.

Given the importance of features and affordances to understand IT-related dynamics, we thus now turn to the features and affordances of microblogging. Microblogging *"allow[s] users to exchange small elements of content such as short sentences, individual images, or video links"* (Kaplan & Haenlein, 2011, p. 106). Twitter (www.Twitter.com) is currently the most popular microblogging platform with more than 100 million users as of March 2012. Twitter allows its users to broadcast freely news and content as well as to share opinions and links to other webpages (Java et al., 2007; Namaan et al., 2011). In times of crises and notable events, between 5,000 and 9,000 tweets per second are posted.[1]

Among the defining features of microblogging is the fact that it only allows for the exchange of extremely short messages or posts. In Twitter, for instance, posts (or tweets) are limited to 140 characters in length. Microbloggers can follow one another on the basis of shared interests by using built-in search and follow functionality. Furthermore, discourse produced through microblogging can be organized as microbloggers assign tags to their posts. In the case of Twitter, in particular, tweeters can attach various "hashtags" ("#") that can help users of the

[1] Source: www.Twitter.com (http://yearinreview.Twitter.com/en/tps.html) (2011 year in review).

service search for microblog posts on specific topics. Moreover, microblogging discourse can be directed as their authors can specify targets for their posts. For instance, in Twitter, directed communications are indicated by the symbol "@."

In terms of affordances, microblogging makes it very easy for people to share links to webpages, sources of information, or other microblog postings, in a variety of messaging formats. The form in which information is disseminated through this particular medium can vary considerably depending on user-related factors that include not just strict familiarity with the medium but also personal perspectives on communicating with others in one-to-many and many-to-many electronic contexts (Bianco, 2009). Furthermore, the ease with which information can be exchanged and organized affords a considerable immediacy (Holotescu & Grossek, 2011) to relay details about quickly developing events which, in turn, could hold diverse consequences according to how particular actors adopt the features available to them.

The richness of the theoretical lenses that we introduced, including NSM theory and multilevel theory, offers considerable insights into the emergence process of activism. More specifically, drawing from NSM theory, we were able to define and contrast individual and collective activism and distinguish between top-down and grassroots activism. Further, building upon multilevel theory, we defined emergence as a bottom-up, dynamic process through which a collective phenomenon appears and gets shaped by lower-level constituents and their interrelations. Much is however still needed to understand how grassroots activism actually emerges through microblogging. The actual process through which microblogging enables, constraints and thus shapes symbolic action in a gradual, bottom-up progression from individual to collective activism is not well conceptualized either. Thus, in order to further our theorization, we now turn to a detailed case study of microblogging in the context of a recent crisis that

triggered activism, the Gulf of Mexico oil spill of 2010. The grounded theorizing that we undertake aims at untangling the emergence of grassroots activism by exploring some of its key constituting dimensions, how they interrelate in this process, as well as how microblogging features and affordances enable this process.

METHODS

Research setting

We chose an opportunistic context, the Gulf of Mexico oil spill (heretofore "Gulf oil spill" or "oil spill"), to develop a grounded theory of emergence. The oil spill (see appendix for more background information) originated from the explosion of Deepwater Horizon, which drilled into the BP-operated Macondo prospect, an oil and gas prospect in the Gulf of Mexico, on April 20th, 2010. The explosion killed 11 men and injured 17 others. The oil spill has caused extensive damage to marine and wildlife habitats as well as the Gulf of Mexico's fishing and tourism industries. The oil spill was, according to statistics released by Twitter[2], the most microblogged about topic of 2010, which revealed wide interests and concerns in the "Twitterverse."

While the oil spill originated from a leak that started in April 2010, the crisis unfolded in a period of several months (until the leak got capped) and its economical, ecological and political repercussions have been ongoing ever since, prompting many activists, in particular environmental activists, to express themselves, organize protests and call for sustainable changes (Casey, 2011; Hoffman & Devereaux Jennings, 2011; Kirsch, 2010; Koger, 2010). The oil spill therefore constitutes an interesting setting for us to examine the emergence of grassroots activism through microblogging over a relatively long period of time (from April 20th, 2010, to November 30th, 2011). It allows us to investigate the initial discourse reacting to the crisis as well as its evolution over time.

[2] http://blog.Twitter.com/2010/12/hindsight2010-top-trends-on-Twitter.html

Data collection

We collected microblog posts published through the Twitter platform. In order to access Twitter's archives we relied upon Topsy (http://topsy.com), a publicly available real-time search engine for the social web. Topsy ranks links, photos and tweets by the number and quality of retweets they receive[3]. The quality of retweets is determined by the "influence" (defined by Topsy's algorithm) of Twitter users (Topsy labs, 2010). Topsy archives tweets and allows users of the service to search for tweets within a defined time period. We used Topsy's Otter API, a RESTful HTTP web service that provides access to Topsy's search results, url information and author information along with the intermediate data (like author influence) that is used in the creation of search rankings.

We first engaged in a pilot data collection by searching for tweets with the hashtag "#bpoilspill" from April 20th 2010 to November 30th, 2011, which resulted in a first sample of more than 5,500 microblog posts deemed the most "important" by Topsy. We then examined the hashtag frequency in this initial sample and discovered that the top three hashtags in our sample were #bpoilspill (by design), #oilspill, and #bp.

Since our objective was to gather a wide range of microblog posts that discussed the oil spill, we extended our data collection to #bpoilspill and #oilspill for the next round of data collection. In order to maximize the amount of data collected from Topsy we searched for tweets containing either #bpoilspill or #oilspill on a weekly basis. This returned up to 1,000 tweets per week while satisfying the API limit of the number of pages accessed daily. Our overall resulting data set contained more than 76,000 tweets (see Table 2).

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[3] http://code.google.com/p/otterapi/

Data analysis

Researchers have become increasingly aware of the opportunities and challenges of large-scale data collection and analysis of social media-based data (Allen et al., 2006; Becker et al., 2011), including for interpretive researchers (Arriaga & Levina, 2010; Vaast & Walsham, 2012). Our goal with this case study was to develop theory through an in-depth exploration of the data in a way that would be inspired by and respectful of the principles of grounded theory (Bryant & Charmaz, 2007; Glaser & Strauss, 1967; Suddaby, 2006; Urquhart et al., 2009).

Many well-cited grounded theory works have investigated relatively "traditional" organizational and IS settings (Galal, 2001; Orlikowski & Robey, 1991; Urquhart, 1999) and relied upon well-known, mostly qualitative data analysis procedures (Charmaz, 2006). Our research built upon large scale and longitudinal data that we hoped were promising for grounded theory building, but that could not readily be handled with the traditional toolbox of the qualitative grounded theorist. We therefore had to be creative to make the meaning of our data emerge in an evocative yet rigorous way (Glaser, 1992). We developed data handling and analyses methods in a highly iterative manner. We went through multiple interconnected moments in our data analyses, including hashtag analysis, word cloud visualization, network analysis and qualitative interpretation of theoretically sampled tweets. Below we provide details about these moments.

To analyze our data, we first aimed at summarizing our data set through descriptive analysis. We thus started by examining the evolution of the most frequent hashtags over our entire observation period (see Figure 1). This Figure illustrated how microbloggers' interests and discourse regarding the oil spill evolved over time.

-- Insert Figure 1 around here --

We then expanded our cross-sample analysis to the actual text of the tweets through a word-association visualization technique named Infomous (http://infomous.com). This visualization technique produces interactive text clouds and can handle very large datasets such as our sample of 76,000+ microblog posts. Developed by Icosystem Corp., the technology parses and summarizes text content and then creates word clouds. The graphical representations seek to highlight the most salient topics emerging from the corpus of data as well as the strength of the relationships between these topics. As such, *"[t]he size of each word reflects the frequency with which it appears in the source, while links between words show proximity in the source."* (Infomous, 2010) In this research, nodes represent the words contained in the content of the tweets, while edges represent the relationship between any two words in the sense that they tend to appear in the same context multiple times (cf. Figures 2-5 and 9-10). The algorithm used by Infomous' cloud diagrams also seeks to establish and represent graphically the *structure* that emerges from a body of text by *clustering* nodes into groups that show a higher density of links within them than between them (Clauset et al., 2004; Newman, 2006). Sometimes referred to as modularity, this network property that sees the division of a network essentially into its component sub-groups, has been investigated in social networks of collaborating researchers (Newman, 2004), in biology in the context of metabolic networks (Guimera & Nunes Amaral, 2005), as well as in online communities (Flake et al., 2002). In this research, the groups correspond to key ideas and concepts that emerged from the ongoing discourse taking place via Twitter around the issue of the oil spill. The word clouds that resulted from this analysis allowed us to identify the most frequent words used in microblog posts at certain moments in time as well as to assess how they were related to each other. Comparing Infomous word clouds over time was also helpful to assess transformations of the collective reactions to the oil spill.

We then looked at how relationships among tweeters formed and morphed during our observation period by focusing on the mention (@) feature of twitter. We modeled the existence of @TweeterB in a tweet by TweeterA as a relationship between TweeterA and TweeterB. Using a social network terminology, we considered two Tweeters to be two nodes in the network, and a Tweet with @ to be a directed edge from the mentioning tweeter to the mentioned tweeter in the tweet.

In order to examine the evolution of relationships among tweeters during the crisis we relied upon social network analysis techniques (Hansen et al., 2010). We performed monthly analyses looking at subsets of targeted tweets per month. We then analyzed the overall network characteristics; first we utilized the average geodesic distance indicator, which is the average shortest path or number of edges between two nodes in the network. This indicator gave us an idea about how dense and connected the network for that month was. Second, we looked at the sub-networks by employing the Newman community analysis method. This allowed us to look at the both the number of communities and their size for each month. Finally, we visualized the monthly networks using the Harel-Koren force based algorithm designed to emphasize the separation between communities and distribute the vertexes equally across the plot area with fewer edge intersection as possible (Harel & Koren, 2002). We qualitatively assessed the differences among the monthly networks visualizations.

Finally, to find more meaning onto tweets related to the oil spill, we qualitatively coded a theoretical sample of our tweets. Guidelines and examples on grounded theorizing and interpretive analysis of electronically mediated settings inspired our qualitative approach (Coleman, 2010; Vaast & Walsham, 2012; Wen et al., 2010). Our coding was especially inspired by the Straussian version of grounded theorizing (Charmaz, 2006; Strauss & Corbin, 1998;

Suddaby, 2006) and combined inductive and deductive aspects. We first identified empirically four dimensions of activism (information sharing, framing, community building, and mobilizing). We then consulted existing literatures on social movements and activism and saw that these dimensions were backed theoretically. We further unpacked these dimensions qualitatively by going back to tweets in order to facilitate the inductive emergence of various instantiations of these roles (seen in rows in italic in Tables 4, 5, 7).

Overall, our research respects the grounded tradition that involves that researchers gradually refine their analyses and interpretations, develop their theory iteratively through multiple back and forth between the data, their analysis, existing theories, and the emerging new theory (Dey, 1999; Gasson, 2003). We triangulated these methods in order to build, refine and constantly challenge our emerging theorizing of emergence. Combining various qualitative and quantitative descriptive tools as well as visualization techniques was essential to our theory building process (Modell, 2009; Shah & Corley, 2006). We stopped this process when our analyses reached a relative stability and our grounded theorizing reached theoretical saturation, i.e. when the addition of new analyses did not drastically change our interpretations (Klein & Myers, 1999; Urquhart et al., 2009; Walsham, 1993).

FINDINGS – MICROBLOGGING THE GULF OIL SPILL

Our data analysis revealed four key dimensions in the emergence of grassroots activism through microblogging in the case of the Gulf oil spill: information sharing, framing, community building, and mobilizing. These dimensions were clearly intertwined but their importance became more prominent at some point or another of the process.

In the following paragraphs, for each of these dimensions, we first provide a definition based on extant literature and then relate them to the emergence of grassroots activism, showing

how each dimension involves symbolic action, focusing particularly on the interplay between discourse and action. Then, we highlight the role of microblogging features and affordances in shaping grassroots activism. The following section will put together these four dimensions and develop a grounded model of the emergence of grassroots activism through microblogging.

Information sharing

Early in the period covered by our data, instances of activism manifested primarily through fragmented information sharing. Information sharing refers to the dissemination of information "*so that it reaches those people to whom it is valuable*" (Malone et al., 1987, p. 390). Initially in the observed process, microblogging was used to share factual content i.e. information that can be verified and that does not explicitly express judgment (Li, 1963) or even "brute facts" (Searle, 1995) i.e. physical things and events that exist independently of human institutions (Agerfalk, 2010; Eriksson & Agerfalk, 2010).

While information sharing occurred throughout our observation period, the exchange of factual information was observed prominently at the beginning. For example, at the onset of the crisis, microblogging posts often involved broadcasting news moment to moment, just as the accident unfolded in the evening of April 20th, 2010 (see Table 3). The information sharing dimension was evident in our analysis, as tweets conveyed various types of information about the oil spill. Some microblog posts, for instance, mentioned direct reactions to the oil spill. Other microblog posts provided details about how BP was responding to the leak or related the U.S. government's reaction to the situation (Table 3).

-- Insert Table 3 around here --

Moreover, our data also reveal that early on, the process was situated at the individual level of analysis. Indeed, as illustrated in Table 3, microblogs were posted by individual entities,

which included individuals as well as individual organizations. Here, microbloggers acted independently and in a fragmented manner; initiatives were disjointed, as illustrated by the fact that the same information was conveyed via different tweets (see Table 3). At this point in the process, words were a critical link to symbolic action and the discursive aspect of activism (Zoller 2005) was salient. According to (Austin, 1962)'s speech act theory, "*to say something is to do something*". Through their posts, microbloggers were "acting;" their "actions" were discursive in nature, done in isolation, in a non-coordinated way.

The role of microblogging: As showed in Table 3, it is interesting to note that information sharing posts often overcame the limitations of microblogging features as a medium (i.e. short messages of 140 characters) by providing links to longer sources of information and evidence. In the same vein, tweeters shortened URLs to convey their messages within the constraints imposed by the technology. Our evidence also suggests that microblogging offered useful affordances for information sharing, as it allowed for people without institutional backing to broadcast news very promptly. Inherent characteristics of microblogging, such as its ease of use, usability and instantaneous nature also contributed to enabling information sharing among users.

Framing

A frame is a schema of interpretation that people rely upon in order to understand and respond to events (Goffman, 1974). Frames help people make sense of the world and act upon it (Davidson, 2002). The process of frame construction, i.e. framing, involves agency (Benford & Snow, 2000): how frames evolve depends upon agents, be they individual activists, mass media, organizations, or more informal groups. Sensemaking is at the heart of the framing dimension of symbolic action. Sensemaking implies inferring a meaningful linkage between symbols and events or experiences (Weick, 1995). It supposes an ongoing process of communication,

interpretation and adaptation. In periods of crises such as that of the Gulf oil spill, sensemaking is critical but challenging as people need to make sense of events that are initially senseless and that shatter existing meanings (Weick, 1993).

Our data set provided multiple instances of framing. For one, framing happened as microblog posts *contrasted the oil spill with other events* (see Table 4). During the early months of our examination period, some microblog posts provided meaning to the unexpected and dramatic crisis by comparing the Gulf oil spill to other, prior and better-understood, events such as terrorist attacks. Comparisons and debates about what the oil spill involved denoted the emergence of a collective sensemaking through microblog posts as their authors jointly discussed and built a sense of what the oil spill meant and of its magnitude in relative terms. Sensemaking also often involved *expressing and eliciting a range of emotions* (Table 4). Such emotions that the oil spill triggered in microbloggers included primarily fear, frustration, anger, sadness and disgust. Framing the Gulf oil spill also involved *discussing its consequences*, including its financial impacts (see Table 4).

The word clouds also provided additional indication of how tweets came to frame the oil spill over time. In particular, contrasting Figures 2 and 3, we note that the label "disaster" did not figure very prominently in the earliest posts (April 2010) but then became a frequent way to characterize the oil spill (from May 2010 on). Moreover, beyond sharing the latest oil spill-related news, tweets increasingly dealt with the implications and impacts of the oil spill and framed them as negative. As showed in Figures 4 and 5 (covering June and October 2010 respectively) considerations related to the consequences on wildlife (with words such as "birds", "turtles", "oiled", "dead") and on health (with words such as "health, "environment" or "science") became prominent over time.

The framing of the oil spill through microblogging also revealed a *tendency toward pointing at "bad guys," or guilty parties* (cf. Table 4). Many microblog postings discussed various parties involved in the oil spill and provided at times scathing critiques of them. The company BP was repeatedly presented as directly responsible for the oil spill. Its CEO at the time of the leak, Tony Hayward, also became a symbol of managerial insensitivity and inadequacy. Over time, criticism went from being directed at only one company, BP, to being generalized to other corporations. Microblog posts at times mentioned the oil spill as an illustration of broader issues especially toward the middle and end of our observation period. The condemnations of certain parties and actors at times turned into expressions of conspiration theories, whereby microblog posts hinted at the oil spill being different from or more than what it seemed and revealed how microblogging practices expressed a criticism and lack of trust toward institutions and large corporations.

-- Insert Table 4 and Figures 2, 3, 4 and 5 around here --

Overall, these examples showed how critical framing is to collective activism as it builds an essential link from the individual to the collective level. Changes in the most commonly used words in tweets related to the oil spill illustrated how individual tweets were becoming congruent and complementary (Snow & Benford, 1988, p. 198) as certain ways of describing and talking about the oil spill were becoming collectively shared.

The role of microblogging: The style and formatting of microblog posts, including the use of capital letters, punctuation and emoticons, played a part in conveying emotions, which in turn, had a substantial role in framing (Table 4). Style and formatting also often conveyed a sense of urgency regarding the information shared (Table 4). Other features of microblogging, such as the "retweet" one, also facilitated and shaped framing. Indeed, the early months of the crisis (April

to July 2010) showed the highest level of retweets taking place among microbloggers who talked about the oil spill. Individuals shared their thoughts and interpretation of events, establishing commonly shared topics and views. These practices contributed to framing as discourse patterns converged around shared sets of expressions and symbols. In this process, the multiplicity of microblog posts and the variety of opinions expressed suggested that microblogging affordances allowed for a deep interweaving of information sharing and sensemaking as tweets were not only relaying "brute facts" (Searle, 1995) but were also, through the sharing of symbolic language, building and broadcasting "institutionalized facts" (Searle, 1995).

Community building

According to NSM theory, a common goal, sense of belonging to a social group, shared history and practices are key to the transition from individual to collective activism (Zoller, 2005). As such, community building is an essential dimension of our process of interest. The building and organizing of communities especially involves a process of identification with an issue and a collective (Ashforth et al., 2008; Lave & Wenger, 1991; Walter, 2004). Identification is inherent to community building as it involves *"an individual's cognitive, moral, and emotional connection with a broader community"* (Polletta & Jasper, 2001) (p. 285). The development of such a collective identity "*refers to positive identification with the welfare of another, such that the other is seen as a cognitive extension of the self, rather than independent*" (Wendt, 1994, p. 384).

As shown in Table 5, community building was especially visible in tweets in which other Tweeters were *mentioned by name.* The oil spill seemed to have been the occasion for rising Twitter celebrities to emerge and some tweeters had become well versed upon who was who and who knew and did what. Moreover, while most microblog postings were undirected, some of

them were *directed at specific Twitter users* and directly addressed them. These instances conveyed closeness among some Twitter users and revealed how information about and framing of the oil spill travelled through emerging Twitter communities via multiple retweets. Retweets and tweets specifically addressed to a particular user also at times made references to offline and not just online relations. Community building also sometimes took the shape of some microbloggers *asking for opinions and reactions* about the oil spill to others in the Twitterverse.

-- Insert Table 5 around here –

Self-selected communities formed and morphed according to the news and their interpretations therefore emerged. Figure 6 reveals how the number of Twitter communities discussing the oil spill, as identified via network modularity analysis (Newman, 2006), changed over time. The number of these communities first increased but then started to decrease over time as they coalesced. At the same time, as shown in Figure 7, these fewer communities were becoming more tightly connected as evidenced by a decreasing *Average Geodesic Distance (AGD)* (*"path distance between two nodes* […] *in other words, the length of the shortest path(s)"* (Yang & Knoke, 2001). The nodes in this particular case are represented by individual tweeters exchanging targeted tweets.

-- Insert Figures 6 and 7 around here –

Considering Figures 6 and 7 together suggests that towards the beginning of the observation period the numerous communities dealing with the oil spill were mostly isolated and disconnected. As time progressed, an increased cohesion emerged whereby previously isolated communities coalesced into larger collectives. A one-tailed *t-test* of the difference in AGD for July 2010 and January 2011 confirmed that, over time, the communities became increasingly more closely connected (*sig*. < .05) (Table 6).

-- Insert Table 6 around here --

A network analysis of the targeted discourse also helps us further analyze evolutions in the patterns of communication. The network diagrams generated on a monthly basis were composed of *nodes*, representing the tweeters who were exchanging information namely the source and the destination tweeter behind a given tweet, and *edges*, accounting for the actual tweet. To get a sense of these changes over time, we contrasted network diagrams from the beginning of the crisis (May 2010), as well as from the months of July 2010 and January 2011 (Figure 8). This Figure provides further evidence of gradual clustering of initially isolated tweeters. At the beginning of the observation period there was a larger number of small, stand-alone communities whose members were not connected to other tweeters (i.e. bottom right quadrant in May 2010 in Figure 8). Over time, though, the number of such isolated tweeters decreased (see bottom right quadrant in January 2011 in Figure 8). Moreover, as time went by more and more tweeters linked to one another, generating communities that were becoming increasingly well-defined with key tweeters enjoying a prominent position of centrality within their respective communities (see July 2010 and January 2011 in Figure XX).

-- Insert Figure 8 around here --

These various sources of evidence illustrate community building as an instantiation of symbolic action. Symbols need a community to be understood and acted upon in a meaningful way (Clark, 1996). Moreover, the identification process through which members of a community come together is built upon and strengthens the collective frames and framing of the community (Taylor & Whittier, 1992), which suggests that framing and community building mutually strengthen each other.

The role of microblogging: Microbloggers relied upon the interactive features of Twitter to share and circulate content that generated a sense of community organized around the topic of the oil spill. The way in which key microblogging features were used thus translated into particular affordances that contributed to community building. Tweets were at times directed at specific users, whereby some users addressed others directly (e.g. via the "@" feature of Twitter). Moreover, tweeters could choose whomever they wished to follow. They therefore self-selected their microblogging community on the basis of shared interests in similar issues. People with no to very weak links came together via shared interests on the Gulf oil spill. The use of the RT feature afforded tweeters the possibility to share content with like-minded others as they spread the word about issues of common concern. Emerging research has supported the idea of community building through microblogging via the development of a collective identity as it has shown that microbloggers are aware of their "imagined" communities (Gruzd et al., 2011). The retweet feature that was key to framing a number of issues related to the Gulf oil spill also played an important role in the building of communities. Sharing and circulating others' posts contributed to community building as it built and expanded upon conversations about the oil spill among people who may not have had the occasion to exchange otherwise (Boyd et al., 2010).

Mobilizing

We analyzed how the framing and community building dimensions developed. Ultimately, these two dimensions unfolded into mobilizing, which involves "*consensus mobilization* (…) *through which a social movement tries to obtain support for its viewpoint*" (Klandermans, 1984), (p. 586) as well as "*action mobilization* (…) [through which] *social movement calls up people to participate*" (Klandermans, 1984, p. 586). Mobilizing is thus an advanced dimension of the emergence of grassroots activism.

Looking at the content of the tweets concerning the Gulf oil spill, mobilizing was at times evident in tweets that *incited "voice"* (Hirschman, 1970) by for instance prompting the audience to put pressure on authorities or to participate in online petitions (see Table 7). Such tweets therefore involved discursive calls for action. The tone of the discourse was often condemning and menacing, as per the example provided (Table 7). Other microblog postings incited concrete actions to respond to the oil spill. Some helped organizing *donations and fundraising effort*; others took the form of *references to online and offline demonstrations* or *encouraged volunteering* to help clean up the Gulf and rescue animals (see Table 7).

-- Insert Table 7 around here --

The word clouds provided additional hints of how mobilizing progressed over time. Right after the accident, some of the most frequent mobilizing words were "cleanup," "stop," "spill" (Figure 3). Later in our observation period, in December 2010, for instance (see Figure 9) the appearance of words such as "claims" or "sues" and in May 2011 (see Figure 10) of "restoration," "green," "energy," "environment," indicated changing calls for mobilization in line with transformations of the collective framing of the oil spill. Here, the clouds revealed that mobilizing was at first mostly dedicated to provide help and relief for the immediate consequences of the spill (cleanup beaches, animals, etc.). Then, mobilizing involved condemning and punishing guilty parties (through lawsuits, fines, etc.) to avoid repeats of the crisis. Finally, mobilizing turned to larger and more abstract issues, such as longer-term promotion of "green energy."

-- Insert Figures 9 and 10 around here --

As Twitter became increasingly used for mobilizing, these calls for collective action built upon and strengthened the sense of community building as well as upon the changing framing of

the oil spill. The emergence of self-selected activist communities was also reinforced by the expression of the first person plural pronoun to engage all involved in these microblogging practices (Table 7). The "we" of such microblog posts assumed a shared community of interests and concerns around ecological issues, and a joint involvement in actions. The use of the "we" also conveyed an expectation that a Twitter-based community might be able to solve the oil spill crisis in conjunction with institutions such as the Environmental Protection Agency (EPA) for example.

A hashtag analysis provided additional insights on mobilizing through microblogging. A particularly interesting example in this regard was that of the "#boycottBP" tag as it suggests a role of microblogging to mobilize a collective toward an activist goal. Our qualitative analysis had revealed recurring calls to boycott BP (Table 7). Moreover, as Figure 11 shows, the use of "#boycottBP" emerged during the very first month of the oil spill, in May 2010, persisted as part of the ongoing discourse and experienced two peaking instances in October and December 2010 before fading into the conversational background. Therefore, from early on during the crisis, "#BoycottBP" helped mobilize people who shared an ambition to act against the company perceived to be at the origin of the leak. Furthermore, as Table 8 shows, during the first peak in "#BoycottBP"-related activity in May 2010, i.e. early on in the crisis, most of the main contributors to this mobilizing were non-institutional actors. This reveals how mobilizing was a grassroots effort that, initially, only peripherally involved existing and established participants such as news media outlets and nonprofit organizations.

-- Insert Figure 11 and Table 8 around here --

The role of microblogging: The formatting of many of our sampled tweets emphasized the expression of resolve and urgency. With hashtags that expressed calls for action, microbloggers

used a specific and new affordance granted to them by microblogging, i.e. the ability to categorize very short messages and to organize strings of such short messages into ongoing, loose conversations. The use of hashtags contributed to the emergence of communities that not only made sense of the oil spill but that were also interested in changing the course of the crisis and in righting wrongdoing. Hashtags often expressed the collective nature of this discursive activism (e.g. "#togetherwecandoanythingandeverything"). Also, calling for retweets was another way of adopting specific features of microblogging for activism. The acronym "RT" (retweet) was frequently seen in microblog posts, indicating that the post was itself a retweet and/or, when preceded by "PLS" [please] that the poster mobilized others to relay the message. Some posts, for instance, jointly revealed an emotional response to the crisis, a desire to act, as well as an awareness of the limitations of the medium for action (Table 7). Hashtags and retweets were also often employed together (Table 7). Finally, another instantiation of mobilizing involved *combining microblogging with other social media (*Table 7), such as Facebook, the video-sharing service YouTube, the joint development of web-apps, fundraising via virtual worlds and crowdsourcing initiatives. This combination of microblogging practices with other social media practices contributed to the emergence of new forms of activism through affordances of social media that allowed people to produce and share discourse as well as to come together and act jointly on the basis of common interests and concerns.

THE EMERGENCE OF GRASSROOTS ACTIVISM THROUGH MICROBLOGGING

We examined how the emergence of grassroots activism was shaped through microblogging in the case of the Gulf of Mexico oil spill. Our data analysis allowed us to uncover a bottom-up process that explains how individual activism morphed into collective activism via information

sharing, framing, community building and mobilizing. In this section, putting together our empirical insights and contrasting them with existing literatures, we develop a grounded theorizing of the process we analyzed, emphasizing its multilevel and dynamic aspects, as well as the progression and relationships among its four dimensions and the roles played by microblogging in the emergence. Figure 12 summarizes our grounded theorizing.

-- Insert Figure 12 around here --

Our evidence showed that the interplay between discourse and action through symbolic action was fundamental in this emergence process. Early on, we observed how activism was enacted through discourse when users were primarily conveying factual information about the oil spill. From a multilevel perspective, activism was at first situated at the individual level. The tweeting entities were at times institutional (e.g. non-profit organizations) and not just individuals, but they were acting independently from one another, essentially in relaying news. Though they were using the same technology and the same platform, they remained autonomous and their stated goals were mainly disjointed. By posting microblogs, tweeters were enacting a form of individual activism that can be labeled *discourse as action* (Austin 1962). However, this discourse as action remained primarily factual as it "*tend*[ed] *to focus on the text itself without explicitly considering context as a crucial resource for interpreting texts as socially and contextually located entities*" (Heracleous and Marshak, 2004 p. 1288).

With time, framing (Benford and Snow, 2000) occurred as microbloggers began to interpret discourse in a more contextualized manner. Here, activism was enacted through *discourse as situated action* (Heracleous and Marshak, 2004) where sensemaking plays a vital part and where the discourse is not only embedded in its context but also nested in its wider social and cultural net. This framing process supposes the occurrence of collective sensemaking

since "*frames are not merely aggregations of individual attitudes and perceptions but also the outcome of negotiating shared meaning*" (Gabson, 1992, p. 111). Sensemaking, which bridges the individual and collective level (Weick 1995), unfolded as microbloggers made sense of discourse through their interactions with others. Initially, as microbloggers struggled to make sense of the information they were receiving, interpretation occurred at an intra-subjective or individual level. As individuals continued to exchange information and expressed how they understood and experienced the oil spill, interpretation increasingly occurred at the inter-subjective or relational level and microbloggers began to construct shared meaning and common frames of reference. Ultimately, a "negotiated order" emerged at the collective level as collective and embedded interpretations of the crisis arose (Drazin et al. 1999). This is consistent with NSM theory that contends that frames develop to achieve certain purposes (e.g. gain more volunteers, acquire resources, change the public opinion) and that framing builds a correspondence between individual and collective identities (Benford & Snow, 2000).

Our analysis of the microblogging of the oil spill crisis revealed that framing was indeed tightly related to the development of collective identities and to community building. A sense of community "*reflects a degree of connectedness among members and a recognition of mutuality of circumstance, including a threshold level of collectively held values, norms and vision*" (McMillian and Chavis 1986 cited in Chaskin 2001). This is in line with NSM theory that has considered "*the construction of collective identities, activist communities, and movement cultures as central categories of analysis*" (Ingalsbee, 1996, p. 272). In the case of the Gulf oil spill, the development of a collective identity in budding Twitter communities was embedded into emerging "collective action frames" (Snow & Benford, 1988) that helped mobilize supporters by

creating a shared sense that the situation was not acceptable and should be changed and that some guilty parties should be confronted.

With community building and framing, bridges between the individual and the collective were created, which in turn provided a foundation for mobilizing. In our case, mobilizing involved calls for actions such as a BP boycott. Framing and community building were intricately related and together helped generate mobilizing. This process is consistent with the view that successful mobilization depends on the construction of collective identities and congruent frames among community members (Ellingson 1995, Snow & Benford, 1988). Hence, grassroots activism is enacted through *discourse as symbolic action* (Heracleous and Marshak 2004) where reality is a social construction. Symbolic language provides a framing for this reality (Phillips and Hardy 2002). Strengthened relations, collective identification and framing in emerging communities (Gould, 1991, Klandermans, 1984) jointly pave the way for mobilization.

Overall, grassroots activism differs from isolated protest events or entities because there is a shared vision, identity, interrelationships, and sense of ongoing action (Zoller, 2005, p. 345). Grassroots activism also differs from more traditional, top-down activism because it does not rely upon an active entity taking the lead on the collective action (Zoller, 2005, p. 345), which makes coordination and motivation to action possibly more difficult. In the case we investigated, microblogging features and affordances made it possible for grassroots activism to emerge through a combination of four related and sequential dimensions of information sharing, framing, sensemaking, and mobilizing that enacted a progression from individual to collective activism. In line with a multilevel theory perspective (Kozlowski & Klein, 2000), individual entities were no longer acting in a fragmented and uncoordinated way but increasingly as a collective exhibiting

converging goals, displaying a sense of membership to a community, and engaged in mobilization.

The role of microblogging features and affordances

Our findings revealed how some microblogging features such as shortened URLs, retweets or hashtags played a vital role in the emergence of grassroots activism. Such features allowed users to share and make sense of information. Microblogging affordances contributed to the emergence of grassroots activism by facilitating the development of a sense of collective identity, which was critical to facilitate users mobilizing around common issues of concern. While it would be ludicrous to claim that grassroots activism is a new phenomenon yielded by microblogging, our study shows how the use of microblogging shaped its emergence in a democratic, comprehensive, cohesive, immediate and dramatic fashion.

Because of its widely accessible, inexpensive and easy to use features, microblogging facilitates *democratic participation* in that it enlarges the array of potential participants and allows equal possibility of contribution from users. Typically, social stratification represents a barrier to full participation in grassroots organizing because it "*insulates community decision makers from change agents*" (Wittig & Schmitz, 1996, p. 57). By allowing users to bypass traditional hierarchies, microblogging transcends the social and spatial boundaries that can inhibit democratic participation. This observation supports early findings on the use of modem links that had indicated that "*electronic networking may obscure social categories, enhance opportunities for innovative group boundaries to form and improve negotiation outcomes*" (Wittig & Schmitz, 1996, p. 58)

Microblogging also enables a multiplicity of expressions that are rich and diverse in nature. Despite the constraints imposed by its features (such as being limited in length to 140

characters), microblogging affordances play a critical role in the *comprehensiveness* of the factual and symbolic information that is exchanged. Despite the multiplicity of information, emotions and interpretations that are shared, our results have indicated that microblogging allows for emerging *cohesiveness* of the flow of information shared and framed. One possible reason is that as communities of users built around a collective identity and share their views through microblogging, others might identify with or join these emerging collectives. A second interpretation, related to the concept of epistemic object may explain the draw of a community. In a nutshell, "*epistemic objects are abstract in nature: they are the objects of inquiry and pursuit*" (Ewestein & Whyte, 2009)(p. 9), which are characterized not only by what they are, but most importantly by the fact that they are incomplete and thus have not yet fulfilled their goal or reached their objective. It might be because of this degree of incompleteness of tweets that additional membership is drawn to the community, from a desire to make a difference, to help shape the outcomes of the mobilization and to achieve its collective goal. In that perspective, our findings provide an alternative view of social movement since previous research had suggested that face-to-face interaction was needed for the integration and development of collective identity in participatory movement organizations (Diani, 2000). Microblogging thus seems to provide an alternative venue for community building and the development of a collective identity.

The *immediate* character of microblogging is an additional factor in this process. Our results indicate that the speed and ongoing flow of information exchange influences both sensemaking and networking. Previous research had similarly found that microblogs are powerful because they provide immediate insight in affective reactions (Jansen et al., 2009) and a fast and easy conduit to notify about events (Holotescu & Grossek, 2011).

Finally, microblogging shapes the emergence of symbolic action in a *dramatic* fashion in that it provides a sense of rapidly unfolding events that would otherwise be impossible to get via more traditional avenues for disseminating information. By creating a sense of urgency, it contributes to motivating users into action. The Internet has been shown to play a similar role in facilitating participation and mobilization in such a way that it can be used as a tactical tool in itself (Pickerill, 2001). Microblogging seems to provide users with a sense of empowerment, a sense that they can ultimately bring about real changes. It thus plays a role in the coalescence, mobilizing and action of activist communities. In an organizational context, previous research had similarly shown that *perceptions* of affordances granted to its users by a particular technology can lead people to change their existing organizing routines (Leonardi, 2011). Our research reveals that this might also happen in the much larger and looser context of microblogging-based activist communities.

IMPLICATIONS AND CONCLUSIONS

Having presented our grounded theory of emergence, we draw implications of this research concerning the significance of grassroots activism through microblogging and the question of the fluidity or persistence of such a phenomenon. We then acknowledge the limitations of our approach before highlighting the conceptual and methodological contributions of our work and suggesting several avenues for future research.

The significance of grassroots activism through microblogging

Microblogging even more so than other social media, including longer-form blogs, makes it easy and convenient for non-institutional actors to broadcast discourse publicly and take part in public discussions over the Internet. Anyone interested can thus participate and become an (individual) activist, in discourse at least. Scholars and essayists have debated the consequences that

microblogging may hold for activism. Gladwell (2010) and Shirky (2011) for instance disagreed upon the meaning and effectiveness of individual activists' engagement through microblogging. Nielsen (2009) also warned against three risks of microblogging for activists: over-communication, miscommunication, and communicative overload. In the same vein, Solow-Niederman (2010) examined microblogging during and after the contested 2009 Iranian presidential election and found that microblogging practices countered the censorship of an authoritarian state, yet failed to assist the citizens of the regime in their actual attempts to mobilize on the ground.

This research adds to these seminal insights through its longitudinal and multilevel perspective on grassroots activism. Of course, tweeters in our case did not need to exhibit the same level of engagement as traditional activists taking to the street, to refer to Gladwell's (2010) critique of social media activism. Our case revealed that some tweeters explicitly recognized the limitations of the new medium and acknowledged that by tweeting about the oil spill they were not necessarily "doing" anything concrete about it. However, our observations also revealed that an unexpected richness and diversity of activist initiatives found a showcase and an outlet through microblogging. Such initiatives included actions that were mere translations onto the Internet of traditional activism (e.g. online demonstrations and fundraisers) as well as more innovative endeavors that took advantage of the affordances of microblogging and of other social media (e.g. crowdsourcing relief efforts, sharing of oil spill-related web apps).

These observations suggest that microblogging might lead to a reshaping of what grassroots activism entails in today and tomorrow's societies. Moreover, our research showed how four key dimensions that connected individual to collective activism through emergence were related and progressed over time, thereby emphasizing the importance of considering

grassroots activism through microblogging in dynamic, rather than only during isolated events or crises.

Fluidity and persistence in emergence through microblogging

A single tweet, restricted to 140 characters, might appear exceedingly abstract and lacking meaning in itself. Emerging research on microblogging has already insisted on the importance of considering a collective corpus of microblog posts to make sense of them and of the events they characterize (Marwick & Boyd, 2010). Context and meaning of microblog posts are thus to be understood through aggregation, which also clarifies the growing influence of viral phenomena such as "Internet memes" (Aral & Walker, 2011; Ratkiewicz et al., 2010; Shifman, 2011).

The multilevel theory-based perspective on emergence that we adopted in this research helped us consider microblog posts as a corpus of changing collective coherence, rather than in isolation from one another. The conceptualization of emergence showed how micro-level constituents come together through compilation or composition and jointly build collective level phenomena. This theorization of emergence can therefore be especially useful to conceptualize and investigate how memes spread via contagion and may contain messages calling for social change (Andrew, 2009; Shirky, 2011).

There might however be questions regarding the persistence over time of social media-enabled phenomena such as the grassroots activism that we examined. Faraj et al.)2011) noted in this regard that the affordances of social media, including of microblogging, can contribute to very fluid communities, i.e. of communities that might form and disband very fast. Ransbotham and Kane (2011) also documented the high turnover of wiki-based communities. Our own research did not directly assess the persistence over time of the membership of these emerging microblogging-based activist communities. However, it showed awareness on the part of

tweeters of the possibly fleeting nature of the content they published through microblogging and their consequential efforts to keep the momentum of their cause alive. It also revealed how mobilizing evolved since the acute and dramatic onset of the oil spill. While at first activism related to the oil spill was mostly dedicated to address the immediate needs associated with the crisis (e.g. cleaning beaches and animals), over time activism took on more long-term and abstract causes (e.g. improving environmental sustainability, developing corporate responsibility). This insight is intriguing because it counterbalances the expectation that microblogging, due to the "immediate" character of the medium, would only be focusing on the latest issues and would lack persistence.

Limitations

Our research had several limitations. In particular, our research examined tweets dedicated to a specific situation, the Gulf of Mexico oil spill. Future research could usefully examine microblogging in the context of other crises. Contrasting our findings with other settings could help enhance the empirical generalizability of our work and develop a more finely attuned understanding of the emergence grassroots activism through microblogging.

Another limitation has to do with the tweets we collected. With more than 76,000 tweets covering a period of over 18 months, this research accessed an interesting and large set of data. However, since we relied upon Tospy, a social media search engine whose algorithm is proprietary and restricts the number of tweets that its users can access on a daily basis, we did not collect the entire population of tweets regarding the oil spill during our investigation period. Neither could we assess how similar the tweets we collected were from the ones we did not collect. Given the large number of collected tweets and given that these tweets were the ones that Topsy's algorithm had deemed the most "influential," we believe that our data set provided us

with a relatively thorough look into microblog posts about the oil spill. Interested researchers might however find value in launching a finer data collection to assess this belief.

Finally, we examined in detail some key features of Twitter (e.g. the "retweet," "hashtag," and "@" ones) because these were the ones that seemed the most centrally relevant to our research question. However, Twitter and other microblogging platforms offer additional features (e.g. promoted trends, Twitter Tales, Twitter widgets, blog links, see (Grace et al., 2010) that we did not analyze. Further data collection and analyses would be needed in order to check if these other features and the affordances they enable are related to the same or different dimensions of activism than we identified.

Contributions

Despite these limitations, this paper held theoretical and methodological contributions. Theoretically, first, this paper built a theory of the emergence of grassroots activism, a form of symbolic action, through microblogging. Doing so, our research contributed to answer Wattal et al.'s (2010) mandate to extend IS research. It also followed Lyytinen and King's (2006) advice for IS research to keep a certain "plasticity" to increase its academic legitimacy and to stay in touch with key trends and transformations in organizational, economical, and societal contexts.

Second, our theorizing contributed to IS research by adopting explicitly a multilevel perspective as advocated in Burton Jones & Gallivan (2007) and illustrated in organizational contexts in Lapointe & Rivard (2005) and Rivard et al. (2011). Our research adds to this perspective by developing a multilevel theory that encompasses emergence at a collective level that is of a much larger scale than organizations or even networks of organizations. Such a large scale is necessary to make sense of new social dynamics associated with social media use.

Third, and finally from a conceptual standpoint, we developed a *process* theorization. We therefore took in consideration not only how the features and affordances of microblogging were intricately related to the four dimensions of activism that we identified, but also, in a more dynamic manner, how these features and affordances combined in time and were related to how these four dimensions and their relations unfolded in dynamic to generate grassroots activism.

From a methodological standpoint this research constituted one of the first attempts to tap into the potential of social media data with the explicit goal of building theory, rather than of exploring data or of testing existing theories. We combined and triangulated methods such as qualitative analysis, world visualization techniques, and network analysis to build a mid-level grounded theory. We hope that other IS researchers will build upon our experience and push this methodological contribution further in order to harness the theory-building potential of these new sources of data and of these contexts of increasing relevance for organizations, businesses, and society.

Future research

In addition to the aforementioned future research prospects, we highlight several promising avenues for research. For one, future research could develop a better understanding of the relationships between online and offline practices for grassroots activists. In a related way, researchers could interestingly examine the relationships between individual, top-down, and grassroots activism through microblogging and other social media. Much remains to be known for instance about how formal activist organizations react to grassroots activism and adopt social media to help pursue their mission.

Finally, we developed a grounded theory of how collective activism emerges from individual activism through microblogging. It would be interesting to test and expand

empirically our theorization by using simulated models inspired by complexity theory that has conceptualized and operationalized emergence (Page, 2010). In this perspective, emergence is a characteristic of complex systems composed of diverse, adaptive, and interconnected entities. Those three characteristics allow complex system to produce dynamic patterns and structures. Running simulations of microblogging networks could help identify and analyze various configurations of emergence.

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TABLES AND FIGURES

Table 1: Comparison between individual, grassroots, and top-down activisms

	Individual activism	**Collective activism**	
		Grassroots activism	**Top-down activism**
Level	Individual	Collective, emerging in a bottom up fashion.	Collective, relying upon leadership of specific entities.
Identification	Identification with an issue (e.g. protection of the environment)	Identification with an issue and development of a collective identity	Identification with an issue and with a formal entity (e.g. an NGO).
Locus of change and methods	Individual and individualized entities, local changes (e.g. green consumer lifestyle), individual symbolic and practical actions.	Community, symbolic and practical actions informally organized not initially recognized by dominant players.	Community, symbolic and practical actions formally organized and recognized by dominant players.
Form of organizing and coordinating	N/A	Informal, bottom-up organizing, emergence of temporary consensus.	Mostly top-down decision-making, possibly bureaucratic organizing.
Agency	Individual	Collective (unspecified)	Collective (specific leaders and formal groups)

Table 2: Number of collected tweets and of tweeters per month

Month	**Tweets**	**Tweeters**
Apr-10 (04/20-04/30)	730	427
May-10	4294	1611
Jun-10	4284	1944
Jul-10	4396	1759
Aug-10	4500	1531
Sep-10	4141	1325
Oct-10	4257	1286
Nov-10	4017	1272
Dec-10	4331	1477
Jan-11	4186	1450
Feb-11	3845	1542
Mar-11	3923	1781
Apr-11	4040	1700
May-11	2749	1258
Jun-11	1967	1016
Jul-11	4239	1916

Month	Tweets	Tweeters
Aug-11	4186	1816
Sep-11	4161	1446
Oct-11	3862	1684
Nov-11	3694	1530

Figure 1: Most frequent hashtags in the collected tweets

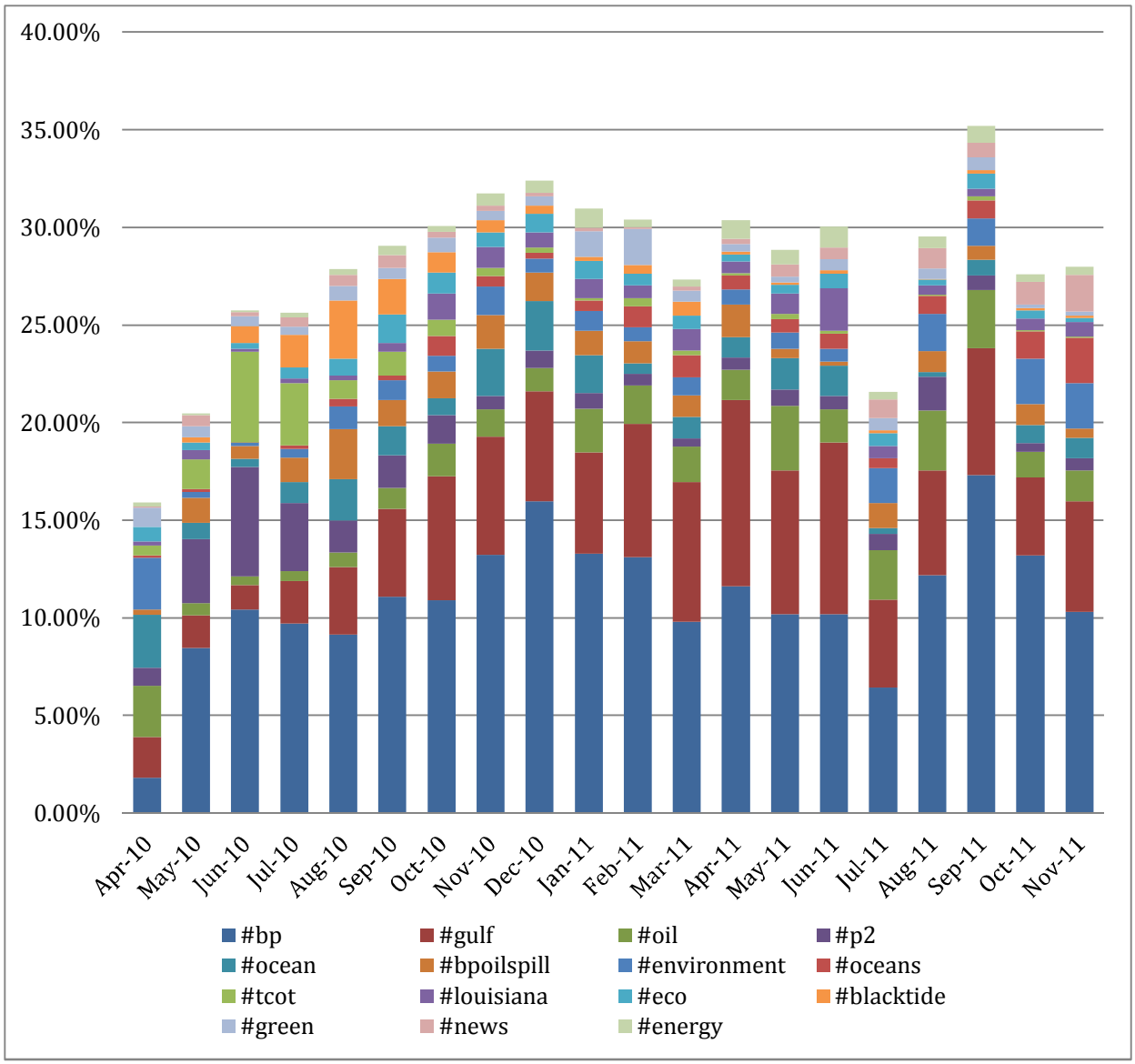


Table 3: Instances of information sharing tweets

Tweeter type	Tweeter	Illustrating Tweet	Date
		Unfolding of events and broadcasting news	
ORG	therightblue	RT @ocean_news: Oil slick measuring 1-by-5 miles spreads from sunken rig, up to 336,000 gallons of oil spill into Gulf #oilspill #ocean	2010-04-22
IND	tkarow	Search 4 #missing #DeepwaterHorizon #oilrig workers 2 B called off soon; rig sinks, concerns abt #oilspill http://bit.ly/d1aBdQ #uscg	2010-04-23
IND	bronxze	Louisiana rig explosion: millions of gallons of oil spill into Gulf of Mexico http://bit.ly/bsxKTU #Oilspill	2010-04-23
ORG	progressflorida	Gulf oil spill was 10 miles wide & 100 square mi. in size this morning. http://bit.ly/bXEGs2 #sayfie #oilspill	2010-04-24
IND	mlavelles	Coast Guard says #oilspill is now 400 square miles. Here's their latest update w/diagrams. http://bit.ly/cPnR7E	2010-04-25
ORG	skytruth	Gulf spill: latest Coast Guard map of spill http://tinyurl.com/2bp76ve #oilspill	2010-04-25
IND	xzendejas	RT @HuffPolitics: AWFUL: Exploded offshore rig spilling 42,000 gallons of oil per day into Gulf of Mexico http://huff.to/9LH0I7 #oilspill	2010-04-25
IND	bigadshaban	Latest on #nola #oilspill Since rig sank in Gulf 4 days ago 168,000 gallons of oil believed to have leaked so far http://tinyurl.com/2ezec4x	2010-04-25
ORG	progressflorida	AP reporting that #oilspill remains possible threat to Florida's shores. #sayfie	2010-04-25
		Information about responses to events	
ORG	bp_america	Live tweeting of Adm. Allen's briefing to begin momentarily. #oilspill #bp	2010-08-16
ORG	bp_america	#BP planning to deep clean Baldwin County beaches soon as tourism season winds down in #Alabama: http://bit.ly/97rO2d #oilspill	2010-10-13
		Same information shared in different ways	
ORG	therightblue	RT @ocean_news: Oil slick measuring 1-by-5 miles spreads from sunken rig, up to 336,000 gallons of oil spill into Gulf #oilspill #ocean	22/04/2010
ORG	ocean_news	RT @nbcnightlynews Oil rig sinks http://bit.ly/aL91WI #oilspill #ocean	22/04/2010
IND	tkarow	Search 4 #missing #DeepwaterHorizon #oilrig workers 2 B called off soon; rig sinks, concerns abt #oilspill http://bit.ly/d1aBdQ #uscg	23/04/2010
IND	greenqueenannie	Major #oilspill as burning #oilrig sinks into sea 11 #people still missing http://tiny.cc/40dnd #oilrig #gulfofmexico #oil	22/04/2010

Figure 2: April 2010 word cloud

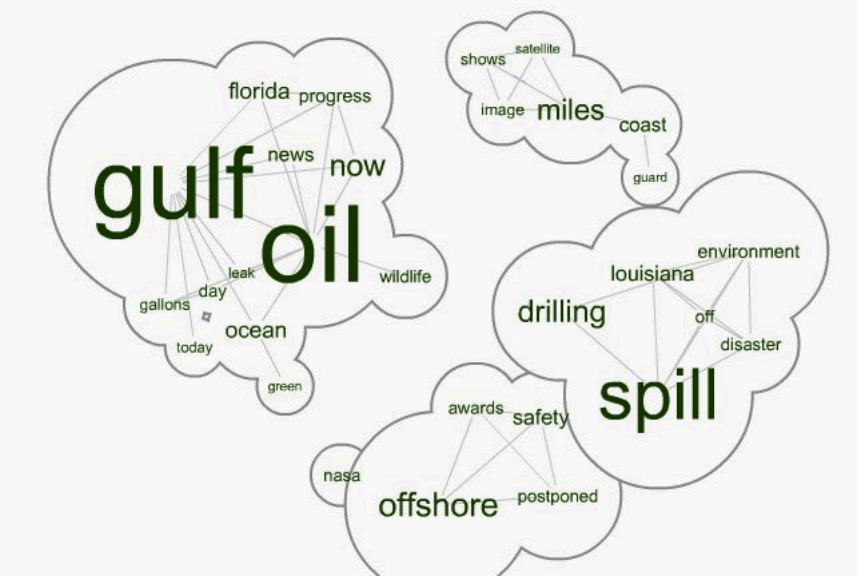


Figure 3: May 2010 word cloud

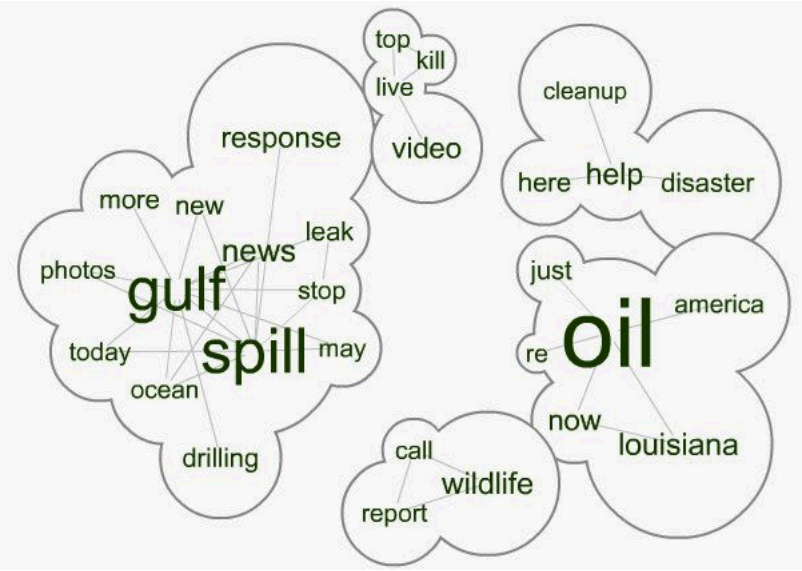


Figure 4: June 2010 word cloud

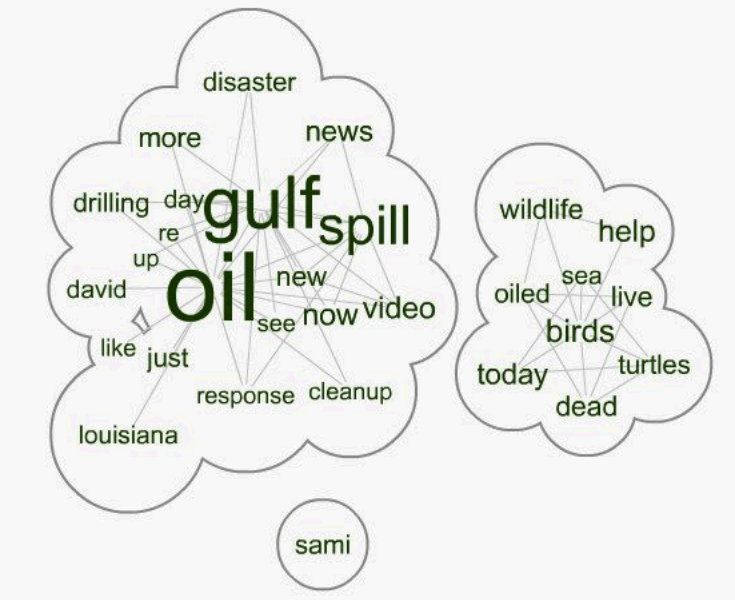


Figure 5: October 2010 word cloud

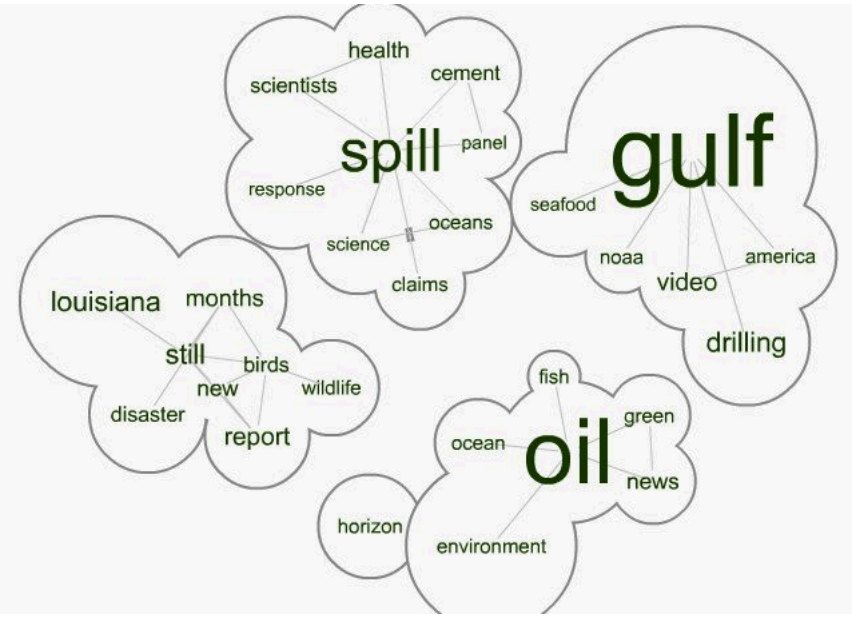


Table 4: Instances of framing tweets

Tweeter	Illustrating Tweet	Date
	Contrasting events	
@waynen tampa	RT @progressflorida: Gulf #oilspill - bigger than the Exxon Valdez (via @SkyTruth): http://bit.ly/9CqpRS #NoDrillFL	28/04/2010
@cleanoc	RT @SkyTruth: Gulf #oilspill: Week 2 begins (Australia blowout went	28/04/2010

Tweeter	Illustrating Tweet	Date
ean	10 weeks). In days this spill may exceed Exxon Valdez - http://tinyurl.com/235x8yd	
@theecofoodie	RT @TheEcofoodie: #Oilspill may take 3months to repair as 200,000gallons leak each day into gulf. Will surpass Exxon Valdez spill--So awful! RT @christackett	30/04/2010
@nihilozero	RT @cureworks: #Oilspill, #Japan #Nuclear #Disaster, etc., the folks responsible are the ones in charge. Fuckery? #change #press #media #BP #prayforjapan	17/03/2011
	Expressing emotions	
whodat35	http://bit.ly/dsCUcZ || Bienvenue/Welcome home Pelicans!!!:) http://bit.ly/d71IBf #oilspill	22/09/2010
greenroofsuk	RT @greenroofsuk: RT @DianeN56: 1st turtles, #BP Burning of Whale: Pls RT this photo!!! http://bit.ly/95eR60 #oilspill RT @_Sea_No_Evil_ RT @StrandedNoMore	05/07/2010
cindyscott54	RT @cindyscott54: #oilspill Obama & Jones Act, you say??? BULLSHIT!! Jones Act is NOT standing in the way of ANYTHING cleanup related. http://bit.ly/9M4yc8	24/06/2010
ariaajaeger	Have a beautiful day everyone, pls keep those prayers going for all the animals and people affected by the #OilSpill #GulfCoast :(	06/06/2010
jerryjamesstone	RT @jerryjamesstone: RT @alexpriest: Think the #BPoilspill is bad? Check out Nigeria... http://bit.ly/cRdYRM // God, SO depressing. :(	18/05/2010
chrisheuer	OMG the drilling mud BP used to try to cap the well last month was toxic with Lye & ethylene :(#bpoilspill http://bit.ly/9B2x5b	27/07/2010
	Discussing consequences	
@orion_magazine	@TOTN on the $ value of a dead pelican: "How can you put a price on millions of years of evolution, of perfection?"#oilspill	21/10/2010
@mawkus	BP tripled ad spending after #oilspill, spent nearly $100 million in just 3 months: http://ow.ly/2y2iM /via @ClaraJeffery @MilesGrant:	01/09/2010
@seachele420	RT @1Sky: $109 billion: Cost of Natural Disasters in 2010 (not to mention the unnatural ones e.g., #oilspill) http://su.pr/1qIQVH via @z ...	02/02/2011
@envirograffiti	RT @biofriendlyblog: #BP #OilSpill Costs to Hit $40 Billion http://su.pr/1XS3vs (via @solveclimate)	03/11/2010
	Identifying guilty parties	
@chrisheuer	stop using #oilspill as hashtag. give credit please, it should be #bpoilspill PLS RT	11/06/2010
@seachele420	RT @greencelebrity: Oil companies must be responsible for their actions http://sns.ly/Ggbqy1 -- #OILSPILL	21/02/2011
@climate_politix	BP Accused Of Trying To Disengage From Oil Spill - Planet Ark: http://t.co/p4RbG0p via @AddThis #BPoilspill	13/07/2011
@sistaseminole	thanks a lot @barackobama renig on thew oil rigs! Oil leaking from blast site http://bit.ly/aTv2A8 // This is bad, very, very bad. #oilspill	25/04/2010
@alanloayza	Hey Palin and Obama, how's that "Drill baby Drill"-y stuff treatin' ya? #oilspill http://huff.to/9rbHlj Huffpost -	25/04/2010

Table 5: Instances of community building tweets

Tweeter	Illustrating tweet	Date
	Mentioning specific tweeters by name	
jwage	Go Leroy Stick! RT @adamtbailey Turns out the guy behind @BPGlobalPR actually knows what he's doing. Awesome. http://ht.ly/1TtiC #bpoilspill	03/06/2010
hettela	I nominate @greenpeaceusa for a Shorty Award in #green because of their #oilspill coverage http://ht.ly/3K59f	27/01/2011
bpgulfleak	Tar balls rolling in http://bit.ly/mmksHL #Pensacola #Florida #oilspill #bp via @stuartsmithlaw (Read this!)	09/06/2011
	Directing tweets at specific Tweeters	
whskr	@winecountrydog Help!! The link fell off. Can I have the link please then I can RT it too! #BPoilspill"	02/08/2010
seasideoil	@BPGulfLeak BP Oil washing in on waves in FL on 9/14/11 #bp #bpoilspill #mc252 #oilleak #bpgulfleak #oildisaster #dwh http://t.co/XcGlsn4P	29/03/2011
davidvitter	@KPLC: Vitter hosting town hall meetings in SWLA http://bit.ly/eUDlLv #oilspill #moratorium #SWLA #Lakecharles	27/04/2011
audubonzoo	@FleurtyGirl: Happy Thankskgiving! Thank you for creating RescueMe tee to help #oilspill turtles. http://bit.ly/seaturtlehope	24/11/2010
	Addressing others in general on the topic of the oil spill	
lornali	Create an oil spill on any web page http://weneed.acleanerfuture.org/ #BPoilspill #eco #green	31/08/2010
brettandersontp	Honestly Someone on Facebook posted this but it is soaked in too much #oilspillirony I had to share #bpoilspill	30/05/2010
rightwingturds	RT @ProgressFlorida: Hey Sens. @MarcoRubio & @MikeHaridopolos - Another #OilSpill, or as you call it a "total anomaly" http://t.co/93zNicm	13/08/2011
	Asking for opinions and reactions	
emmergroup	We're not so far from the #Gulf- are you worried about your #seafood? http://ow.ly/5mKYX #BPoilspill #aftermath	22/06/2011
bpgulfleak	RT @SeaWeb_org: Powerful collection of voices from the #Gulf Coast. What are your hopes for the future?http://ow.ly/4M21h #oilspill #BP	03/05/2011
nwf	Do you think loggerhead sea turtles should be listed as endangered? If so please take action: http://bit.ly/aI8Krk #oilspill #DT @NWF	23/08/2010
kjr8	RT @zobba_com: How long do you think we will be seeing the effects of the oil spill? http://bit.ly/g5QyEz #oilspill	19/01/2011

Figure 6: Number of Twitter-based communities talking about oil spill

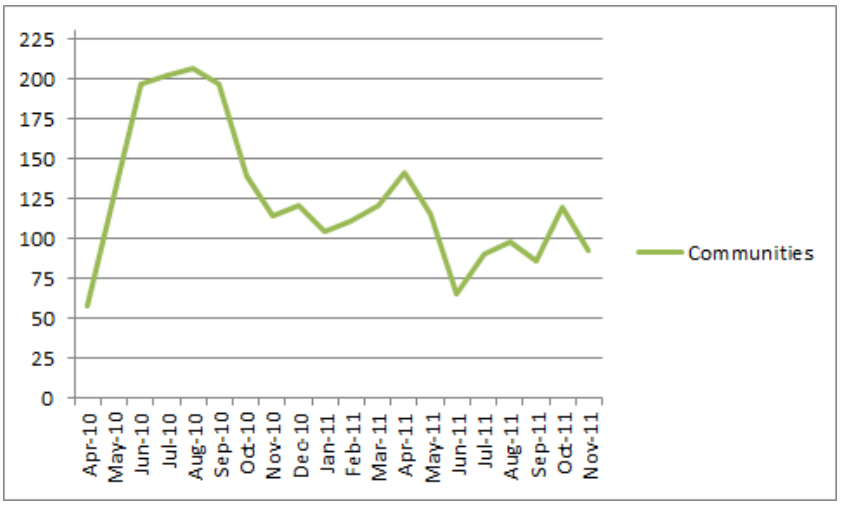


Figure 7: Average Geodesic distance of Twitter-based communities talking about oil spill

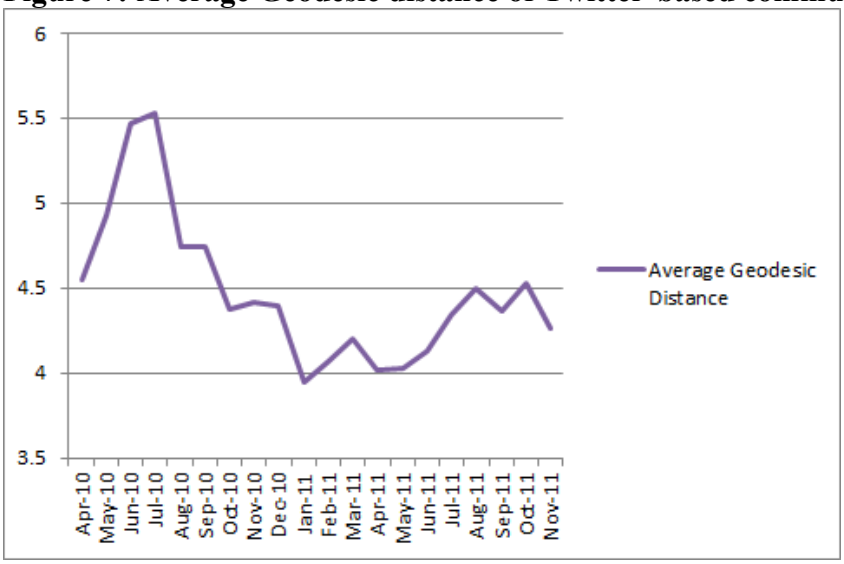


Table 6: One-tailed t test of AGD between July 2010 and January 2011

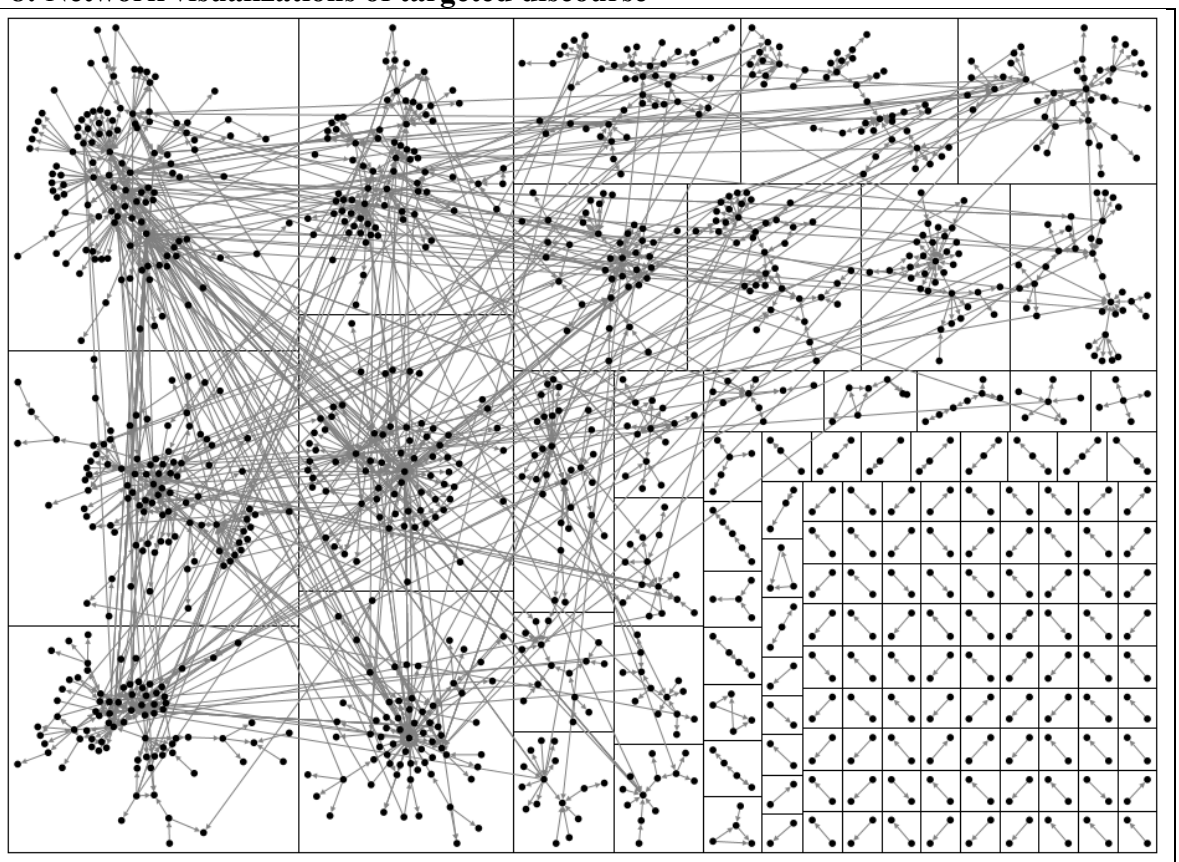
Two-sample t test with equal variances

	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
x	1803	5.34	.4474616	19	4.462402	6.217598
y	2545	3.95	.3766255	19	3.211476	4.688524
combined	4348	4.526396	.2882975	19.01016	3.961186	5.091606
diff		1.39	.5848663		.2433637	2.536636

diff = mean(x) - mean(y) t = 2.3766
Ho: diff = 0 degrees of freedom = 4346

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.9912 Pr(|T| > |t|) = **0.0175** Pr(T > t) = 0.0088

Figure 8: Network visualizations of targeted discourse

May 10	

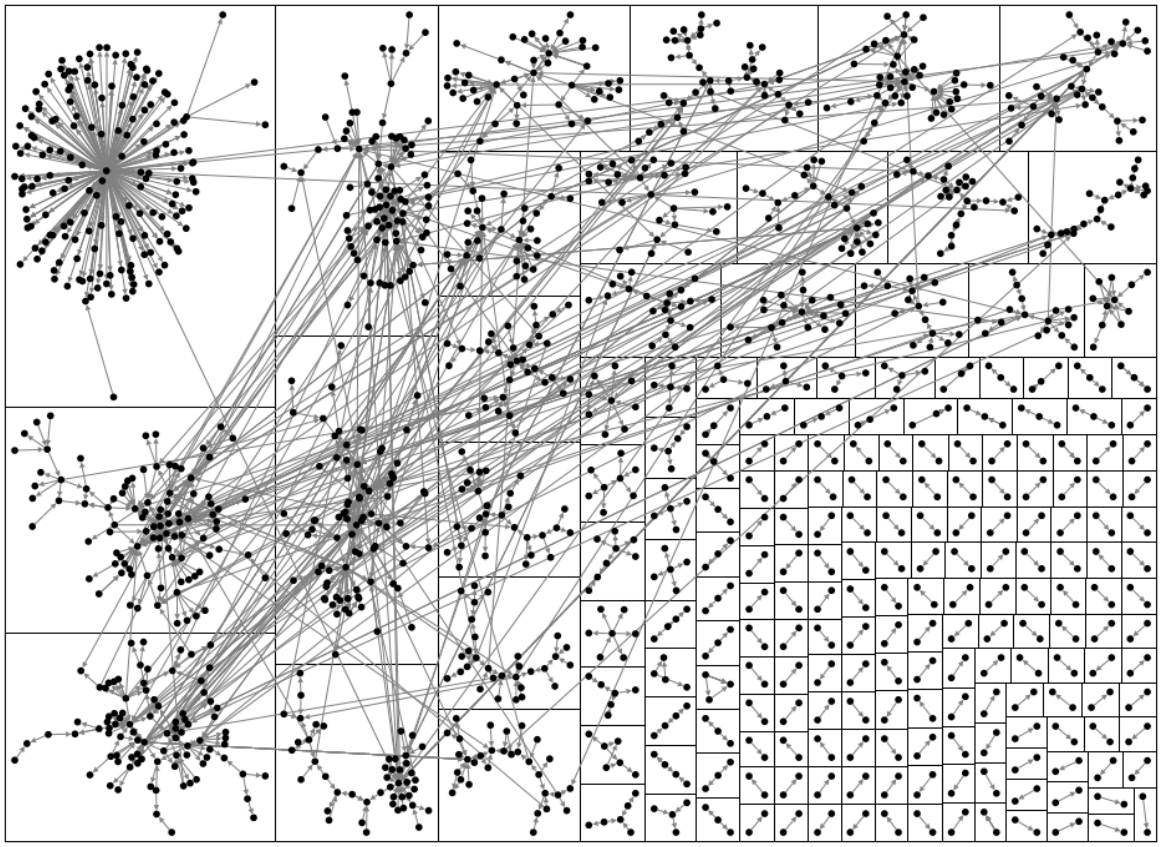
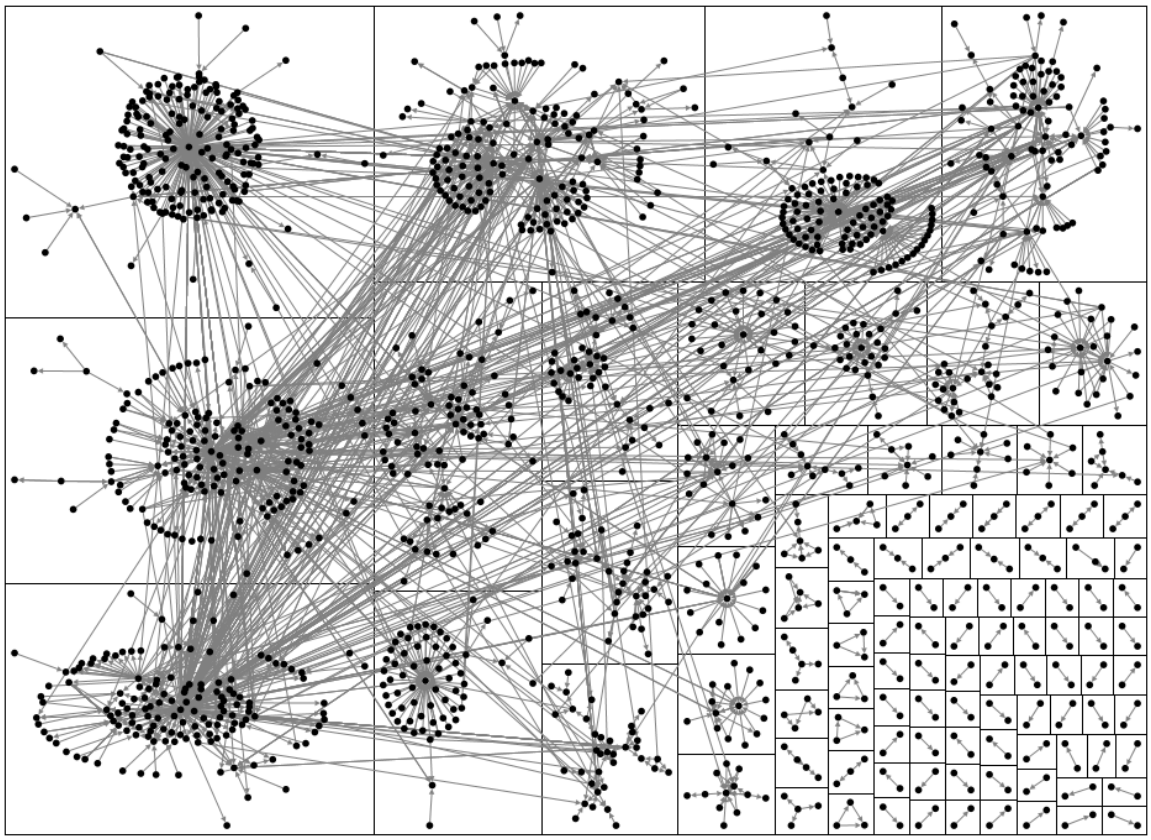
Jul 10	
Jan 11	

Table 7: Instances of mobilizing tweets

Tweeter	Illustrating tweet	Date
	Inciting voice	
joebert72	Contact your senators RIGHT NOW & tell them to support the RESTORE Act http://t.co/T1lDOER - hold BP accountable! @HealthyGulf #BPOilSpill.	26/07/2011
beachnriver	Sign this Letter to #BigOil to stop them putting profits before planet, people and economy! http://bit.ly/blzmgU #oilspill #bpoilspill.	06/11/2010
barbiesnow	RT @MoveOn: ACT NOW: Corporate Corruption like the #OilSpill is NOT acceptable. Help us end it: http://j.mp/Stand4Dem #Stand4Dem #Other98	21/06/2010
	Organizing donations and fundraising efforts	
oilspillcleanup	#bpoilspill relief sponsored by Pay4Tweet. $2 will give you a tweet and support orgs working on the cleanup - http://ow.ly/1Fry2.	25/07/2010
autoday	Ideas people, help @TheGulfCafe get donations to http://t.co/QAYSHKw - A quick survey at http://t.co/WDJY7tk #oilspill #blacktide #seafood	17/07/2011
rachelannyes	RT @ThePacificOcean: @rachelannyes NGOs are rounding up volunteers, supplies. Ppl can help spread the word or w/ fundraising http://bit.ly/a4smVD #oilspill	30/04/2010
audubonzoo	Donate $5 to help #oilspill sea turtles. Macy's will match your donation & you'll get special shopping discount! http://bit.ly/9ZoY8A	26/08/2010
	Discussing if tweeting meant "doing" something	
mirelamonte	As heart wrenching as these photos are, pls [please] RT. I wish there was more being done to help. http://bit.ly/c2jdm6 #BPOilSpill #oilspill	11/11/2010
whodat35	@BP_America. is that tweet for real? #oilspill Don't drag our Mardi Gras into your propaganda. We'll party when #Gulf fully restored. #NOLA	12/02/2011
treehugger	RT @HollandAC when tweeting ab #oilspill hold #BP accountable! 20 yrs later Exxon brand still = spill, but #BP is staying out of headlines	01/05/2010
	Referring to online and offline demonstrations	
mrremains	10,000 Students Rally in DC Calling for Action on #Climate and #BPOilSpill @TreeHugger http://t.co/fVcNsEL #powershift	15/06/2011
yesmagazine	Hands Across The Sand; Acts Against Offshore Drilling With Global Demonstration http://huff.to/bQbyKF #p2 #BPoilspill #gulf.	18/07/2011
ras_menelik	RT @BPGulfLeak: Anti Dispersant Demonstration - Tampa convention center Oct 9 #nalco #corexit #bp #oilspill http://bit.ly/a1LZp7	05/10/2010
	Encouraging volunteering	
puppynumber7	#URGENT Can you pls [please] help 51 #Dogs in Danger effected by #BPOilSpill http://tl.gd/9qdhmg This is a desperate #NoKill	24/04/2011
earl_piggot	A Universal Message From Twitter Users - Clean The Gulf http://youtu.be/L17KGETfUls #gulfspill #green #oilspill #BP #BPOilSpill #BPOil	25/06/2010
colormered	RT @britishpollute: Help Volunteer In The Gulf	11/06/2010

Tweeter	Illustrating tweet	Date
	Combining microblogging with other social media	
angelfrmcanada	RT @georgek1029: @MMonides @EileenLeft Think U have a solution 2 the #BPOilSpill? File it here https://gw.innocentive.com/ar/challenge/9383447 Pls [please] RT	01/07/2010
bizofdiving	rt @OpSqueegee July 1-15 #SecondLife fundraiser for Nat[iona]l Wildlife Fed #NWF Call for Events, DJs, Bloggers http://bit.ly/9ES7K9 #BPOilSpill	24/10/2010
joebert72	Get Involved! The @HealthyGulf BP Oil Spill action website. Log onto http://bit.ly/cazbAf #Bpoilspill	26/07/2011
	Encouraging boycotts and changes of practices	
merlyn43	"#Boycott #BP" Don't buy #gas from #BPoilspill, BP taking shortcuts & wrecking the #environment http://t.co/4OITRqA2 #GulfOilSpill #Gulfspill	15/09/2011
progressflorida	Thinking we need to start supporting companies that create renewable energy like @magnegas #oilspill #bpoilspill #green http://ow.ly/1U7ok	29/07/2010
katieemo	Did U [you] know U [you] could Submit y[ou]r CLEAN UP the Gulf #BPOilSpill 2 the EPA [Environmental Protection Agency]? We c[a]n solve th[i]s problem http://dld.bz/hwFk #Ingenuity	07/05/2010
markmayhew	RT @SpillBabySpill: If You're Not Outraged, You're Not Paying Attention! ===BOYCOTT BP NOW=== @Greenpeace @greenpeaceuk @greenpeaceusa #oilspill	21/05/2010
bpgulfleak	Facts regarding BP and their treatment of our citizens and environment since capping well = even more reason 2 BOYCOTT #oilspill #Corexit	15/11/2011

Figure 9: December 2010 word cloud

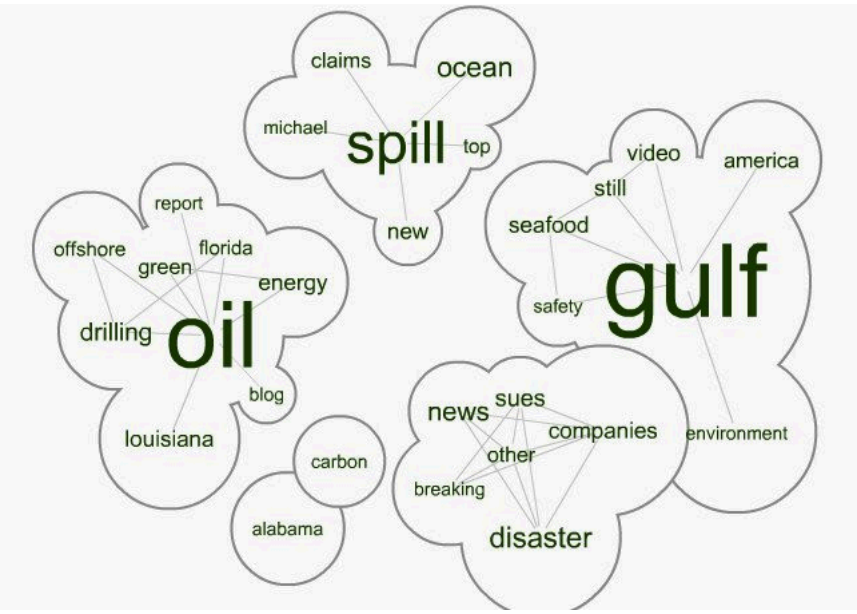


Figure 10: May 2011 word cloud

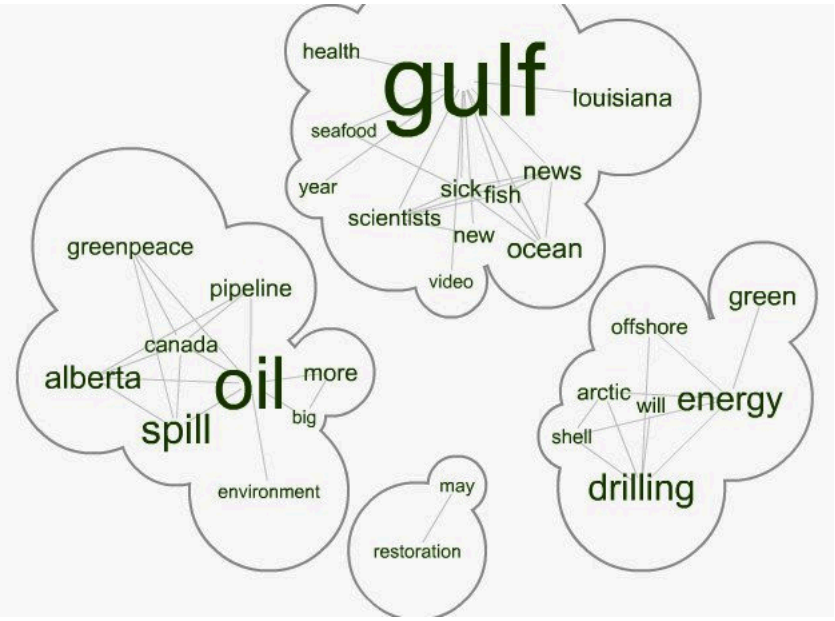


Figure 11: Number of tweets calling for #BoycottBP

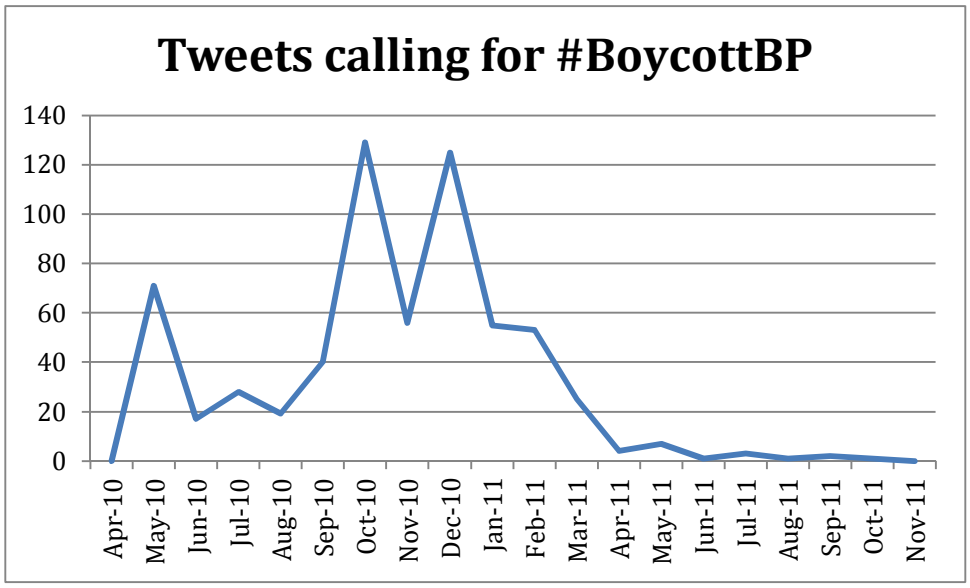
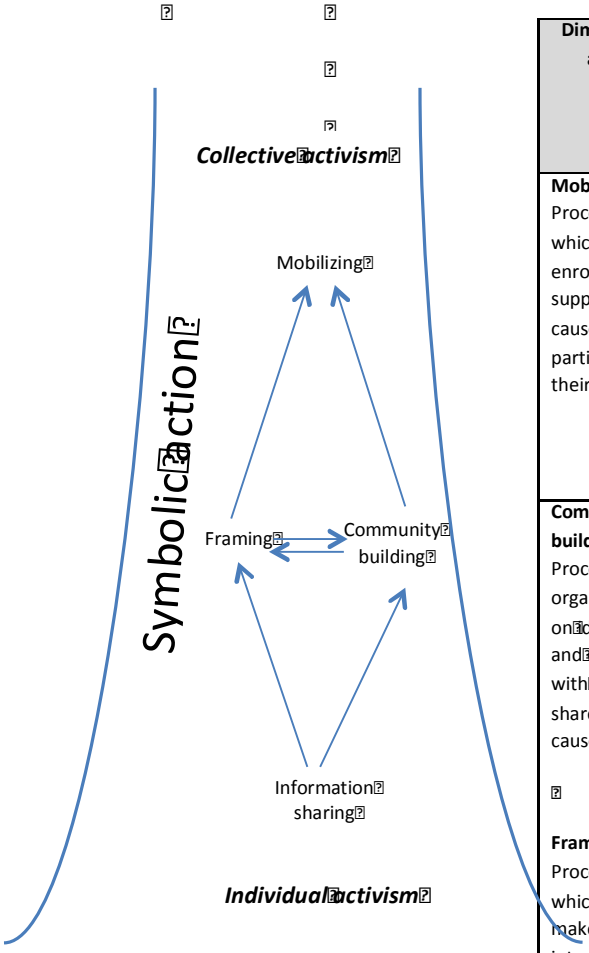


Table 8: Top users of #boycottbp in their posts (May 2010)

Hashtag	Institutional actor	Non-institutional actor
BoycottBP	@hmbascom @NatGeoSociety	@NOLAGaYToR @Deighardt @canjcat @CabaretArtist @cindybax @gingerkiller @csammells @mcoop @JuliaAdair @Ann7837 @noupsell @DianeN56 @TheRagingCelt @sandro

Figure 12: A grounded theory of emergence of grassroots activism through microblogging



Dimensions of activism	Expressions of activism	Key microblogging features	Key Microblogging affordances
Mobilizing: Process through which activists enroll others to support their cause and participate in their actions	Practical language --**Discourse** as *symbolic action*	#feature External links to social media	Proving a platform for calls for action Putting together meaning and action via hashtag aggregation Avenue for coming together: acting jointly
Community building: Process of organizing based on identifying and connecting with others who share a common cause **Framing**: Process through which agents make sense, interpret and respond to events	• Symbolic language (collective identity) • Symbolic language (sense making) --**Discourse** as *situated action*	@feature RT feature Style and formatting	Multiplicity of expression of opinion, emotions and interpretation Building sense of community by sharing issues of common concern
Information sharing: Dissemination of factual information	Factual language (brute facts) --**Discourse** as *action*	140-char limit Shortened URLs	Event broadcasting in a timely fashion Non-institutional, democratic participation

APPENDIX: BACKGROUND ON GULF OIL SPILL

The Gulf of Mexico oil spill (aka Deepwater Horizon oil spill, or BP oil spill), heretofore "oil spill" involved the release of a liquid petroleum hydrocarbon into the environment that took place in the Gulf of Mexico. Liquid petroleum hydrocarbon flowed for three months from April to July 2010 and, as of November 2011, was still leaking oil[4]. The oil spill is the largest accidental marine oil spill in the history of the petroleum industry[5]. The spill started from the explosion of Deepwater Horizon, which drilled into the BP-operated Macondo prospect, an oil and gas prospect in the Gulf of Mexico. The explosion killed 11 men and injured 17 others. Capping the gushing wellhead mostly stopped on July 15th, but the wellhead had already released about 4.9 million barrels (780,000 m^3) of oil[6]. In August 2011, oil covering several square miles of water surfaced near the well but was too dispersed to be recovered.[7] The oil spill caused extensive damage to marine and wildlife habitats as well as the Gulf of Mexico's fishing and tourism industries. In November 2010, 4,200 square miles of the Gulf were closed to shrimping due to tar balls in shrimping nets. In November 2010, as well, 320 miles of Louisiana shoreline were affected by oil [8].

Oil spill timeline[9]

April 20, 2010 – Gas, oil and concrete from the *Deepwater Horizon* exploded up the wellbore onto the deck and then caught fire. The explosion killed 11 platform workers and seriously injured 17 others.

April 22, 2010 – Rig sinked. Oil leaking leaking at the rate of about 8,000 barrels (340,000 US gallons; 1,300 cubic metres) of crude per day.

April 25, 2010 – Oil sheen seen covering 580 square miles (1,500 km^2) and is 70 miles (110 km) south of Mississippi and Alabama coastlines and was 31 miles (50 km) from the ecologically sensitive Chandeleur Islands. BP begins process to establish two relief wells.

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[4] Sources: Wikipedia entry "Deepwater Horizon oil spill" (http://en.wikipedia.org/wiki/Deepwater_Horizon_oil_spill), retrieved 12/18/2011 and Market Watch, Wall Street Journal, Is BP'S Macondo Reservoir Leaking More Oil?, Nov. 6th, 2011, http://www.marketwatch.com/story/is-bps-macondo-reservoir-leaking-more-oil-2011-11-06, retrieved 12/18/2011

[5] Sources, Wikipedia entry "Deepwater Horizon oil spill," BP leak the world's worst accidental oil spill," *Telegraph*, 08/03/2010, http://www.telegraph.co.uk/finance/newsbysector/energy/oilandgas/7924009/BP-leak-the-worlds-worst-accidental-oil-spill.html. Retrieved 12/18/2011, Gulf Spill Is the Largest of Its Kind, Scientists Say, *New York Times*, 08/02/2010, http://www.nytimes.com/2010/08/03/us/03spill.html?_r=1&fta=y , retrieved 12/18/2011

[6] Sources: Wikipedia entry, "Deepwater Horizon oil spill," PBS' NewsHour, New Estimate Puts Gulf Oil Leak at 205 Million Gallons, 08/02/2010, http://www.pbs.org/newshour/rundown/2010/08/new-estimate-puts-oil-leak-at-49-million-barrels.html retrieved 12/18/2011

[7] Sources: Wikipedia entry, "Deepwater Horizon oil spill," NOLA, *Times-Picayune*, Coast Guard says oil sheen 16 miles northeast of BP well too dispersed to be recovered, 09/01/2011, http://www.nola.com/news/gulf-oil-spill/index.ssf/2011/09/coast_guard_says_oil_sheen_16.html, retrieved 12/18/2011

[8] Sources: Wikipedia entry "Deepwater Horizon oil spill," Potential Impact of the Gulf Oil Spill on Tourism, Oxford Economics, 2010, http://www.ustravel.org/sites/default/files/page/2009/11/Gulf_Oil_Spill_Analysis_Oxford_Economics_710.pdf Retrieved 12/18/2011.

[9] Sources: Wikipedia entry, "Deepwater Horizon oil spill", ibid, and Wikipedia entry Timeline of the Deepwater Horizon oil spill, available at: http://en.wikipedia.org/wiki/Timeline_of_the_Deepwater_Horizon_oil_spill retrieved on 12/18/2011.

April 28, 2010 – The National Oceanic and Atmospheric Administration estimated that the leak was likely 5,000 barrels (210,000 US gallons; 790 cubic metres) a day, five times larger than initially estimated by BP. BP announced controlled test to burn oil off the surface had been successful.

April 29, 2010 – Louisiana Governor Bobby Jindal declared a state of emergency.

April 30, 2010 – Oil washed ashore at Venice, Louisiana. President Barack Obama halted new offshore drilling unless safeguards were in place.

May 5, 2010 – BP announced that the smallest of three known leaks had been capped allowing the repair group to focus their efforts on the remaining leaks.

May 7, 2010 – A 125-tonne (280,000 lbs.) container dome lowered over the largest of the well leaked and piped the oil to a storage vessel on the surface.

May 8, 2010 – BP reported that methane was freezing at the top of the dome making it ineffective.

May 9, 2010 – Tar balls reported on Dauphin Island in Alabama.

May 10, 2010 – After failed containment dome BP announced plans to apply five feet in diameter containment vessel nicknamed "top hat". BP announced strategy of trying to push mud and debris down the tube to clog it. The strategy was nicknamed "junk shot."

May 11, 2010 – BP, Transocean and Halliburton officials testified before Congress blaming each other for the incident.

May 12, 2010 – BP released first public video of leak.

May 13, 2010 – BP CEO Tony Hayward called the oil spill "relatively tiny" in comparison with the size of the "ocean."

May 15, 2010 – Coast Guard and EPA authorized use of dispersants underwater, at the source of the Deepwater Horizon leak.

May 19, 2010 – Oil washed ashore on mainland Louisiana.

May 21, 2010 – BP begun live underwater video broadcasts of the leak.

May 22, 2010 – Obama signed an executive order establishing the bipartisan National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling.

May 26, 2010 – BP announced plan to force feed heavy drilling mud in a project called "top kill".

May 29, 2010 – BP declared Top Kill a failure and moved on to their next contingency option, the Lower Marine Riser Package (LMRP) Cap Containment System.

June 4, 2010 – Tar balls arrived on beaches in Pensacola, Florida.

June 2010 - BP set up a $20 billion fund to compensate victims of the oil spill.

June 25, 2010 – Hurricane Alex (2010) caused relief rigs to disconnect and let the oil spill unchecked into the ocean.

July 10, 2010 – Old cap removed from well at 12:37 p.m. CDT in preparation for a new cap.

July 12, 2010 – Three ram capping stack installed on the Deep Water Horizon. The stack completed the installation of the new 40-ton containment device sealing cap.

July 27, 2010 – BP board formally announced that Bob Dudley would replace Tony Hayward as BP CEO as of October 1, 2010.

December 15, 2010 – The U.S. federal government sued BP Exploration and Production, Inc., and eight other corporations, for unlimited liability; it also sought civil penalties under the Clean Water Act.

January 2011 – White House oil spill commission released its final report on the causes of the oil spill.

January 13, 2011 – An oil spill commissioner reported that tar balls continued to wash up, oil sheen trails were seen in the wake of fishing boats.

February 26, 2011 – A research team found oil on the bottom of the seafloor that did not seem to be degrading.

May 26, 2011 – The Louisiana Department of Environmental Quality extended the state of emergency related to the oil spill.

As of July, 2011 – Roughly 491 miles of coastline in Louisiana, Mississippi, Alabama and Florida remained contaminated by BP oil, according to a NOAA spokesperson.

As of July 2011 – BP compensation fund had paid $4.7 billion to 198,475 claimants. In all, the fund had nearly 1 million claims and received thousands of claims weekly.

September 2011 – The U.S. government published its final investigative report on the accident. Report stated that the main cause was the defective cement job, and Halliburton, BP and Transocean were, in different ways, responsible for the accident.

October 2011 – NOAA reported dolphins and whales continued to die at twice the normal rate.